



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATURAL SCIENCES YEAR 2 TERM 1 TO 4 TEACHER'S GUIDE



Term 1: LIFE AND LIVING

Content overview

	Year 2	Year 3	Year 4
LIFE AND LIVING	<u>Living and non-living things</u> Living things • <u>Identify</u> and explain • Briefly explain the seven life processes • Non-living • Identify • Explain why things are non-living Practical: Living and Non-living things	<u>Digestive system</u> • Explain concepts • Describe structure • Name the parts and the organs • Describe the features and functions of the parts and organs	<u>Sense Organs</u> Senses • Briefly define concepts • Name and describe animal senses adapted for survival
	<u>Structure of plants</u> • Describe the basic structure of plants • Identify differences between plants	<u>Nutrients in food</u> Food Groups • Briefly define concepts • Categorise food • Describe the functions of food groups	<u>Human skeleton</u> • Briefly define concepts • Function of bones
	<u>Structure of animals</u> • Describe the basic structure of animals • Identify differences between animals	<u>Nutrition</u> • Explain a balanced diet	<u>Human Reproductive System</u> Reproduction • Introduce and explain concepts with examples • Explain the physical changes • Explain the features in the human organisms
	<u>What plants need to grow</u> • Explain requirements to growth • Explain how plants grow from cuttings and seeds	<u>Food processing</u> • Explain concepts, importance of preserving, types of products, methods and benefits	

Content outline

WEEK	TOPIC	CONTENT
1-3	Living and Non-living things that share the world with us	Living things - Identify and explain different kinds of living things - Explain all seven life processes: feeding, growing, reproducing, respirating, excreting, sensing, and moving carried out by living plants and animals - Explain with examples why some things appear to be dead but will come alive, given the right conditions, e.g. dried yeast, and dried beans, an incubated fertilized egg. Explain why some things were living and are now dead: e.g. dead wood/leaves. Non-living things - Identify and explain different kinds of non-living things: - Explain why some things were living and are now dead: e.g. dead wood, dry leaves. Practical activity: Identify and explain the seven life processes of any living organism.
4-5	Structure of plants	Features of plants - Describe the following concepts with examples: roots, stems, leaves, flowers, fruits, seeds. - Describe the basic structure of plants: roots, stems, leaves, flowers, fruits, seeds - Identify visible differences and similarities between plants: e.g. small, large, grow in water or soil, produce fruits, flowers; thick/thin/large/small leaves etc. Practical demonstration: Learners must touch, smell and observe the differences in plant structures.
6	What plants need to grow	Requirements for growth of plants - Explain the requirements for plants to grow: light, warmth, water, gases and fertile soil. - Explain and demonstrate how new plants can be grown from cuttings or seeds. Practical Investigation: Growth of plants from seeds and cuttings.

7-8	Structure of animals	Structure of Animals • Explain the following concepts with examples: Head, Tail, Body, Limbs, Sense organs, Body covering, Vertebrate, Invertebrate Features of animals • Describe the basic structure of animals: head, tail, body, limbs, sense organs, body covering, and vertebrate, invertebrate. • Identify visible differences between vertebrate and invertebrate animals.
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % Test (Pen and Paper): 50 %		

Topic 1: Living and non-living things: week 1-3

KEY QUESTIONS

- What does it mean to be alive?
- What is a non-living thing?
- What does it mean to be non-living?
- A chicken egg seems to be non-living, but then it can hatch into a chicken. Is the egg living or non-living?

Introducing the topic

Teachers are encouraged to help learners with language-related problems by providing a vocabulary rich environment and to display key words in a prominent place while constantly referring to these words. This supports learners to learn new scientific vocabulary and use it to explain themselves.

In this unit, the learners will find out about what all living things have in common. The emphasis must not be on memorising facts or definitions but on activities that use process skills such as: **observing differences, sorting and classifying, describing and drawing.**

As a teacher, you need to build the language needed for talking about concepts. Even if the learners use the correct word, the meanings of that word may be different for each of them. It is especially important to introduce the formal scientific terms of the seven life processes correctly and to explain each process and word-meaning accurately. Although more technical terms may be introduced incidentally, CAPS requires that the seven life processes are named as follows: **moving, reproducing, sensing, feeding, respirating, excreting and growth.**

Learners need to understand the seven life processes and distinguishing between living and non-living things. It is suggested that teachers make a display with visuals of these words: cut coloured A4 pages lengthwise in half, write or print the process on each long strip and use prestik to attach it to the wall in a mind-map formation. In the centre of the mind-map write: "The Seven Life Processes". As examples of each of these processes are studied in the class let the learners add illustrations or interesting facts and build the mind-map as you work through the section.

Teachers Note: It is important to understand that Living things (Biotic) are small units that make up these living things. It can die. All living things are either 'dead or living'. A dead animal or plant is still classified as a Living thing.

Non-living things (Abiotic) do not have small units that are living and are immortal (never dying or decaying)

There is a wide variety of living things, including plants, animals and fungi (mould, yeast, amongst other).

1.1. LIVING THINGS

There is a wide variety of livings, including plants and animals. They are distinguished from non-living things by their ability to move, reproduce and are sensitive towards their environment. To survive they need water, food, a way of getting rid of waste and an environment which stays within a particular range of temperature.

INSTRUCTIONS

1. Look through these photos and decide whether you think they are living or non-living.
2. When you are done, discuss your choices with your class.



Flowers and plants



Water and waves ¹



A zebra



Clouds in the sky ²



Chicken eggs in a nest ³



A burning fire ⁴



A tropical fish ⁵



Mould growing on a lemon ⁶

There are things that look as if they are non-living, which are living given the right conditions for example seeds, yeast and mould.

1. Introduce the unit with a class discussion. Encourage all learners to participate by calling on different learners for their opinions on what constitutes being living and non-living. Use the white or black board and write down the key concepts that are raised.
2. Ask the learners to name the living things around you and at home.
3. Let the learners examine the photos to identify the seven life processes. Let them explain each in their own words. Encourage learners to remember the seven life processes. For example:

M = Movement

R = Reproducing

S = Sensing

R= Respiration

F = Feeding

E = Excreting

G = Growth

Characteristics of living things

Living things may look different, but they all carry out seven life processes.

Let's take a look at each of these.

MOVING: All living plants and animals move

- Humans and animals use their bodies to move from one place to another.
- Some plants turn towards light or water. Roots mostly grow downwards. Many stems grow upwards.



Humans move all the time. These athletes are running.

REPRODUCTION: All living things reproduce (babies or seeds)

- Humans and animals have babies.
- Some new plants can grow from seeds.
- Other plants grow from cuttings or shoots.



A mother and father with their baby

SENSING: All living things respond to any change that they sense

- When you are feeling cold, you will put on a jersey or jacket.
- When it becomes winter some animals hibernate.
- In autumn the leaves on some trees change colour.
- You can use an umbrella to protect you from the rain or from the harsh sun on a hot day.
- Reptiles like to lie and bask in the sun on cold winter's days.



This chameleon is basking on the wall in the winter sun.



The leaves on some trees respond to the change of the season. These leaves are turning brown during autumn.

RESPIRATION: All living things exchange gases in and out

- Humans and animals use the gas, oxygen from the air that they **breathe** in.
- They release (give off) the gas carbon dioxide when they breathe out.
- During respiration plants take in the gas carbon dioxide into their leaves. They use it to make food. They then release oxygen for animals and humans to use.

FEEDING: All living things need food

- Food gives all living things the energy they need.
- Green plants can make their own food for energy in their leaves and stems.



Picture retrieved from: [\(1293\) Pinterest](#)

EXCRETING: All living animals and plants have to get rid of waste products

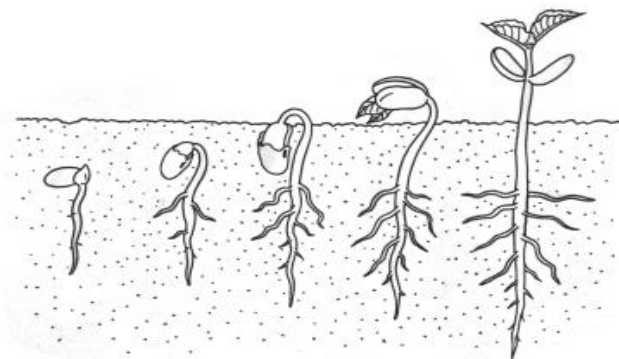
- Humans and animals have to get rid of waste products from their bodies.
- Plants get rid of waste water through the process of transpiration.



Do you see how shiny the horse looks? She is sweating from all that running! ⁹

GROWING: All living things grow

- Human and animal babies grow into adults.
- Seedlings grow into plants.



Growth of a seed into a seedling

All living things carry out seven life processes. Non-living things do not carry out any life process.

Teacher's Presentation hints

1. In the next activity, learners are going to check their understanding of living and non-living things by doing the activity based on the seven life processes.
2. You could demonstrate. Every learner has to complete the activity in his/her workbook.
3. Go through the answers

Summary

Life processes

Living things carry out **seven life processes**. They **feed, breathe, sense, grow, produce, excrete** and **move**.

Life Process	Description
Feeding	All living things feed.
Growth	All living things grow.
Reproduction	All living things reproduce.
Respiration	All living things take in oxygen to release energy from food.
Excreting	All living things get rid of waste.
Sensing	All living things respond to changes in the environment.
Movement	All living things move.

ACTIVITY: Understanding the seven life processes

INSTRUCTIONS:

1. Look carefully at each object.
2. Decide whether the object is **living** or **non-living** and write your answer in the middle column.
3. Provide a reason for your answer in the last column.

Object	Living or non-living?	Reason
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 ¹² An aeroplane	non-living	No life processes
 ¹³ Fish in the sea	living	Have life processes
 ¹⁴ A plant	living	Have life processes
 ¹⁵ A bouncing soccer ball	non-living	No life processes

	living	Have life processes when provided with the right conditions
¹⁶ Chicken eggs		

Informal assessment by teacher

Learners mime (act without using words) the life processes.

Learners write down the seven life processes. (Use acronym: Mrs R Feg.)

Ask informal questions:

1. How does the movement of plants and animals differ?
 - Movement - animals can move from one place to another, while plants grow in the same place but can move towards the light or towards/away from gravity.
2. What are the ways that we can use to grow new plants?
 - seeds, cuttings
3. Are the cooked vegetables living or non-living? How do you determine this?
 - A good way to explain this is to ask if they could plant the cooked vegetable and if new plants will grow from it. If they cannot then the item is no longer living

Some things seem to be non-living, but they are living!

Yes, some things seem to be non-living for a very long time. When exposed to the right conditions they revive again, they show the characteristics of living things.

Look at the picture below of seeds which seem to be non-living!



Picture retrieved from: [Dry Bean Seed - AGT Foods](#)

QUESTIONS

1. Why do seeds seem to be non-living? How can we show that they are living?

Learners might think seeds are non-living as they do not appear to show any of the seven life processes. But things can appear to be non-living when they are actually in a dormant state. You can germinate the seeds to show that they are living.

We say the seeds are in a "dormant state" until they are given water, warmth, air and light to germinate and start growing. There are other things too which seem to be non-living, but if they are given the right conditions then they can revive and carry on living.



A dove keeping her eggs warm to hatch them.



[How to hatch eggs at home without incubator \(and with one\) in 5 Easy Steps \(piwakawakavalley.co.nz\)](http://piwakawakavalley.co.nz)

Fertilised eggs need to be kept warm or they will not hatch. This is why a mother bird will start sitting on her eggs to keep them warm when she has laid all of them.



Retrieved from: [How Yeast Is Used in Cooking \(thespruceeats.com\)](http://thespruceeats.com) ; [What every new baker should know about the yeast all around us \(theconversation.com\)](http://theconversation.com)

Yeast can cause bread dough or cake to raise.

ACTIVITY: Can I revive living things that seem to be non-living?

MATERIALS (what you need):

Photos of eggs hatching

INSTRUCTIONS:

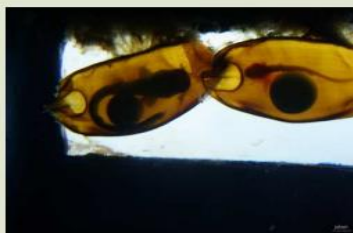
1. Look at these photos carefully.



Chicken eggs hatching. ¹⁹



Frog eggs about to hatch into tadpoles. ²⁰



Shark eggs. ²¹



Snake eggs hatching. ²²

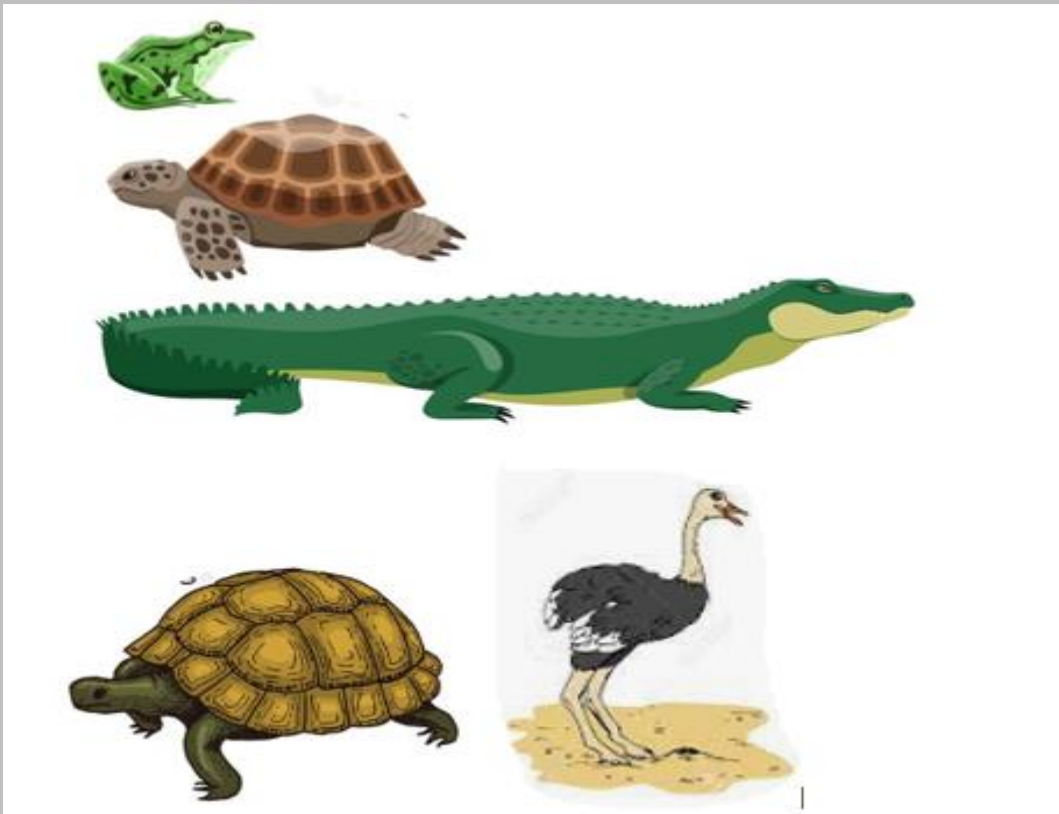
QUESTIONS:

1. Study the photo of the bird sitting on her nest. Can you explain why she needs to sit on her eggs?
 - **She needs to keep the eggs warm in order to hatch them.**
2. Farmers often do not let the mother chickens sit on their eggs. Instead they put the eggs in something called an incubator, as you see in the picture above of the chicken eggs hatching. What does the incubator provide to the eggs?
 - **It provides warmth.**

3. A snake normally lays her eggs in a 'nest'. Why does she not have to lie on top of them to hatch them?
- Snakes are cold-blooded, so even if the snake did lie on top of the eggs (like birds do) it would not help them hatch as her body would not provide any heat. The heat of the sun provides the warmth to hatch the eggs.
4. Have you ever caught tadpoles or kept silkworms in a box?
- Maybe someone in your class has some that they can bring to school. In what season can you normally find little tadpoles or silkworms, and why? Silkworms are typically available in spring and early summer, and tadpoles can generally be found in pools and ponds from spring to autumn. The eggs need warmth to hatch. The adult animals lay their eggs in these seasons because the warmth of the sun will hatch the eggs.

5. RESEARCH QUESTION

South Africa has a wide variety of animal laying eggs.



How do different animals hatch their eggs?

Frog turtle/tortoise crocodile ostrich

Your answer needs to include the following information:

1. Where do they lay the eggs?
2. When do they lay the eggs?
3. How do they hatch?

4. How long does it take for the eggs to hatch?

5. Provide pictures.

In your class group discuss all the similarities and differences between the ways in which the animals hatch their eggs.

Possible Answers

Animal	Research Information
Frog 	Where: Female frogs usually lay them in water to keep them moist after a male has fertilized them How: Most hatch into fish-like tadpoles, complete with gills and a tail. How long: Frog eggs take anywhere from three days to (25days) three weeks to hatch
Turtle  How do sea turtles hatch? (noaa.gov)	Where: All turtles, including sea turtles, lay their eggs on land . Female turtles dig their nests in summer. How: After laying eggs in the hole and covering them with dirt, the female departs. How long: Turtle eggs take about 2 to 3 months to hatch.
Crocodile 	Where: Crocodiles lay eggs close to bodies of water. How: Crocodiles are either hole nesters or mound nesters, i.e. they bury their eggs , or they construct a mound , usually out of vegetation, and deposit their eggs in the centre of it . How long: Crocodile eggs take approximately 9–10 weeks to hatch. Crocodiles lay eggs once a year .
Ostrich	Where: Eggs are laid in a communal nest called a dump nest



How:

How long: Ostriches do not lay eggs throughout the year like chickens do.

They have a **specific breeding season** that starts in June/July every year.

The birds will lay one egg every second day.

ACTIVITY 3: Germinating a seed (To show that a seed is a living thing)

MATERIALS: (what you need):

- bean seeds
- cotton wool
- plastic lids (from empty peanut butter jars for example)

Bean seeds are the most common to get hold of, but sometimes take a bit long to grow. Lentils or coriander seeds can sometimes grow quicker. Perhaps try a mixture of the seeds.

INSTRUCTIONS:

1. Place two layers of cotton wool in the plastic lid.
2. Place a few of your seeds between the two layers of cotton
3. Drizzle water over the seeds. You need to water the cotton wool enough to wet it but NOT TO DROWN THE SEEDS! There should be NO WATER running over the sides of the lid or your seed will drown!
4. Place your seeds in a warm place near a window.
5. Water your seeds whenever you feel the cotton wool is almost dry.

Be careful not to drown your seeds!

QUESTIONS:

1. Before you water your seeds, describe how they look and feel.

Learners should refer to the hard and dry texture of the seeds and that it looks non-living.

2. Draw your seeds between the cotton wool on the first day.

3. Check the progress of your seeds every day. How long did it take them to germinate? And what do your seeds look and feel like now?

Learners should refer to the outer layer of the seeds getting soft and moist and say that there is something growing out of the seeds.

4. What do you think made your seeds revive?

The water that was poured over the cotton wool kept the seeds moist (as if it was lying in moist ground). Some might also remember that they put the seeds near the window so the heat from the sun might have also helped the seeds to sprout. This is discussed extensively in the later section so do not spend too much time on it.

ACTIVITY: DEMONSTRATION (Getting yeast to grow!)

MATERIALS (what you need):

- packet of dry yeast
- sugar
- warm water
- an empty yogurt tub or any transparent container

INSTRUCTIONS:

1. Place a teaspoon of sugar and a teaspoon of dry yeast in your yogurt tub. Mix with your spoon.
2. Add 3 teaspoons of warm water.
3. Stir your sugar and yeast mixture in the warm water to make sure it is well mixed.
4. Watch to see what will happen!

SAFETY WARNING! Don't use boiling water - it might burn you! Boiling water will also kill the yeast. You only need to use lukewarm water!

QUESTIONS:

1. What does your yeast look and feel like before you mix it with the sugar and water?

The yeast feels like small round balls / grainy / dry / grey / non-living.

2. When you add the sugar to the yeast, does anything change in the yeast?

Not really - it stays the same.

3. What happened to the yeast and sugar mixture when you added the warm water?

It starts to bubble and smells 'strange'. The mixture bubbles up into the tub. There is a faint fizzy noise.

4. How did the yeast revive?

The yeast needed the warm water and the sugar to revive from a dormant state.

1.2. NON-LIVING THINGS

Non-living things are different from living things because they do not perform seven life processes.

Introducing the next activity

Non-living things do not ever perform any life process

1. The teacher provides the learners with pictures to identify whether these pictures indicate if things are living or non-living, if they are non-living and can become living or if they were once living but are not non-living.
2. Pictures could also be from newspapers or magazines showing living or non-living objects, toys, bones, yogurt, dried seeds as lentils, beans, flower bulbs, electric bulbs, etc. Challenge learners to bring a variety of objects.
3. Divide the learners in groups before they engage in the activity below. Tell them that they have 10 minutes to complete the activity. Instruct them to work together, to follow the instructions and to complete the activity. Walk to each group and help the groups that are unsure.
4. Ask the group presenters to report back on what they learnt. Groups must be quiet during the report session and listen to one another.

Discuss a few examples and allocate time for learners to ask questions. Let every learner complete the table in his/her workbook using the objects their group brought to class. Encourage them to add a few more examples to their table

ACTIVITY: Distinguishing between living and non-living things

The reason for doing this activity:

- To help you learn to distinguish between living and non-living things.

MATERIALS (what you need):

- 3 - 5 different pictures with objects that are living or non-living (bring them to school)
- Scrap paper or cardboard

INSTRUCTIONS:

1. Use the scrap paper or cardboard to make four labels of the following headings:
 - Living
 - Non-living
2. Show the pictures or objects you brought to your group. Place each item or picture under one of the headings you just made.
3. Now complete the table below with the results. If there is time left over you can add in interesting objects from other groups into your table as well.

These answers will depend on the objects brought by the pupil

Living	Once lived	Can be revived	Never lived

4. Carefully look at these photos. Say which of these are living or non-living, or which was once living or can be revived. Write these labels below each of these.

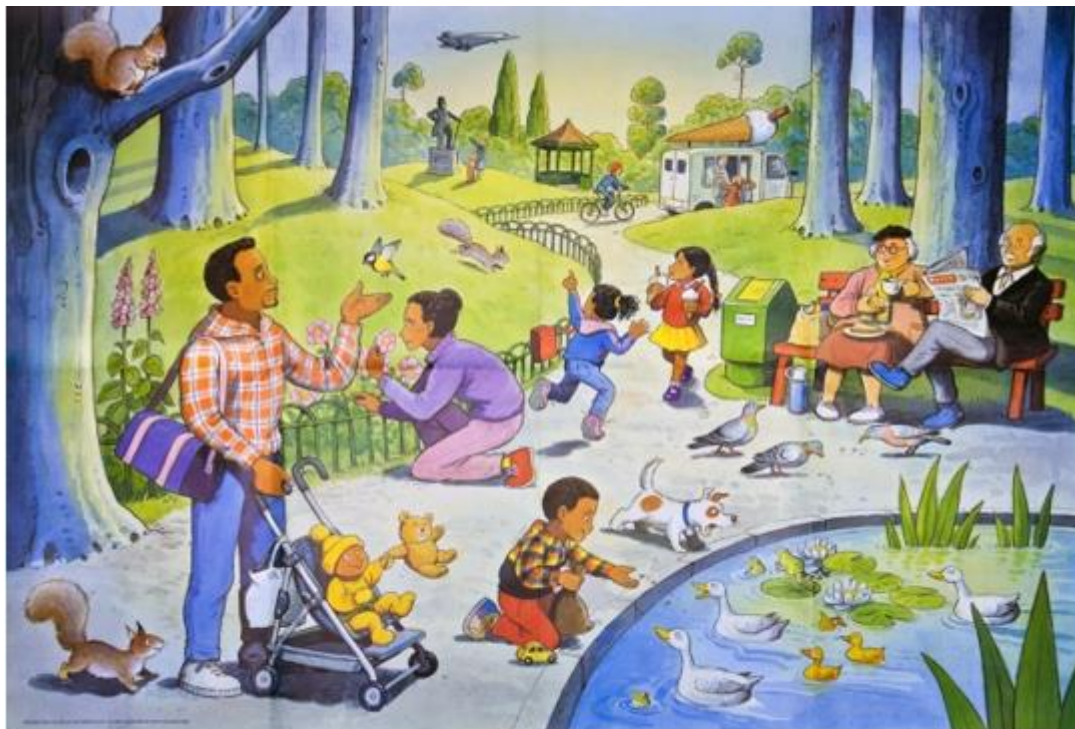
 A dog ²⁴	 Traffic lights ²⁵	 Eggs ²⁶²⁷
Living	Non-Living	Can be revived
 Fire ²⁸	 Trees ²⁹	 Paper
Non-living	Living	Once living
 A dolphin ³⁰	 A computer ³¹	 A skull ³²
Living	Non-living	Once living

KEY CONCEPTS

We can group things on Earth as living and non-living.

- There are seven life processes that all living things carry out.
- Non-living things cannot carry out seven life processes.
- Living things can die.
- Some things like seeds or eggs can seem to be non-living but they are living

REVISION



Picture retrieved from: [What living and non living things can you spot in this picture? | Kingfishers' Class Blog \(wordpress.com\)](http://www.kingfishersclassblog.wordpress.com)

QUESTIONS

1. Name five non-living things you can identify in the picture.

Any five toys, bench, newspaper, paving, shoes, water, clothes, bicycle, etc. (any relevant answer)

2. Name five things in the picture that use oxygen.

People/boy/man/ladies, trees, plants, ducks, squirrel, dog

3. What life process in living things uses oxygen?

Respiration

4. Give an example from the picture of:

- a) moving: child running/ bird flying
- b) sensing: lady smelling the flower
- c) feeding: the ducks eating/ old lady drinking tea/ child eating ice- cream
- d) growth: the green leaves / baby/boy/girl /older people- **different stages** of human life, ducklings

5. What life process do we associate with the eggs in the nest? **Reproduction**

TOPIC 2: STRUCTURE OF PLANTS: WEEK 4-5

KEY QUESTIONS

- Are all plants made in the same way with the same parts?
- If I cannot see leaves on a cactus, is it still a plant?
- Is the moss that grows near an outside tap a plant? How can I tell if it is?
- If a cactus, seaweed and a dandelion are all so different, how can you say they are all plants?

We learnt that almost everything can be grouped into two groups - living and non-living. In the next section we are going to learn more about living plants and animals, how they look and what makes them special!

BASIC STRUCTURES OF PLANTS

Note to the teacher

Teachers are encouraged to help learners with language-related problems by providing a vocabulary rich environment and to display key words in a prominent place while constantly referring to these words. This supports learners to learn new scientific vocabulary and use it to explain themselves.

Introducing this topic

- Learners need to identify different basic parts of plants.
- They need to explore VISIBLE differences between different plants.

The following information is important to teach to the learners.

Basic structures that applies to all plants

The basic parts must be introduced with functions:

Stem, roots and leaves are the basic parts of all plants.

Plants can have flowers, fruits and seeds.

Learners need to observe differences and find out more about the different types of plants in their environment. The differences can be the size of plants, the colour and form of the leaves and flowers, whether they have flowers or not, whether they lose their leaves or not - and many similar visible differences. We encourage teachers to ignite learners' curiosity and interest to find out how the natural world works. As young scientists, they need to compare plants or animals and group them together.

1. Always try to link learners' prior knowledge to what they are going to learn before you formally present the lesson.
2. Bring a plant or a clear poster showing the basic structures of flowering plants to the class.
3. Ask learners to identify different parts of the plant. Encourage them to think of the possible function(s) of each part.

4. Use questions to guide the class discussion regarding the basic structures of flowering plants. Why do plants need water? Which part of the plant can take up water? Give another word for "take up"(absorb). Which part of the plant connects the roots to the rest of the plant etc.?
5. Explain new words clearly and carefully and make sure these words (and possibly their meanings) are displayed in a prominent place in the class.
6. Encourage learners to set up their own glossary with explanations of the meanings.

All plants have the following basic parts:

- roots
- stems
- leaves

Let's take a look at the different (parts of the plant

Roots

Plant roots are normally found underground. Roots have very important functions:

- roots anchor the plant in the ground
- roots absorb water and nutrients from the soil, which are then transported to the rest of the plant
- some plants store the food they make in their roots, like carrots. Later on you will see how plants make their own food!



The roots of this tree go deep down into the soil.



A carrot is a root that stores food made by the plant

Picture of tree: [Are Tree Roots Messing with Your Home's Mojo? - King Piers LLC](#)

Stems

Stems connect the roots to the rest of the plant. The stem has important functions:

- the stem supports the leaves, flowers and fruit

- the stem carries nutrients and water from the roots to the other parts of the plant
- some plants store the food they produce in their stems (like sugar cane or asparagus or potato)



A growing plant stem.¹



A tomato plant stem.

Leaves

Although many plants' leaves are green, leaves can have many other colours. Some leaves change colour during autumn. Leaves have the following functions:

- leaves absorb sunlight and use it to make food for the plant.

Note to the teacher! This process is known as photosynthesis - learners are however not required to learn such specifics at this stage

- some leaves store water.
- most leaves have veins which are like tiny pipes that carry water and nutrients from the roots (the veins carry the food the leaf makes to the rest of the plant)



Can you see the veins in these leaves? ²

Stem, roots and leaves are the basic parts of all plants. Plants **can** have flowers, fruits and seeds.

In the following activity learners will be required to label the drawing. This is the first time that the skill of labelling is being introduced in Year 2, so it is important that teachers actively teach learners this important skill step by step, waiting for

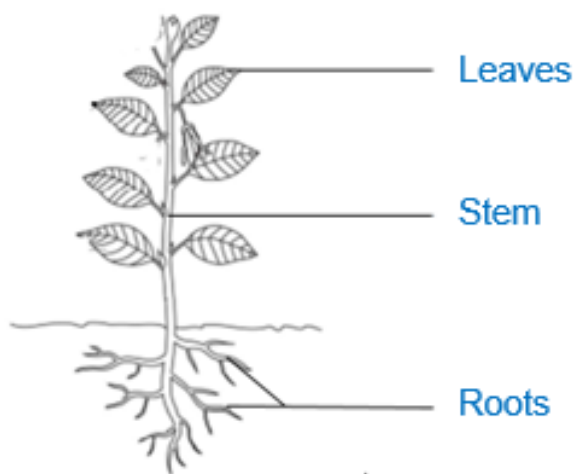
everyone to finish one step correctly before moving onto the next step. Perhaps the following rules could be made into a poster to be displayed and referred to whenever learners are asked to make scientific drawings....

The guidelines for drawing and labelling:

- The drawing must have a heading (printed in pen).
- Both the drawing and the labelling lines must be in pencil.
- Labelling lines must be drawn using a ruler.
- Label lines must be parallel to the top / bottom of the page.
- Label lines must touch the part of the drawing being labelled.
- Label lines must end the same distance from the drawing (i.e. the labels must be in a vertical line underneath each other).
- Labels must be written using print, and not cursive, and in pen.
- The correct labels must be used in the correct place.

Label the structure of the plant below.

Answer



Visible differences between plants

This section on plants provides an opportunity and a necessity to practise agricultural science as a part of Natural Science. For example, while doing this section, encourage your learners to plant different kinds of crops (those that store food in stems, leaves, in roots, etc.) and then ask them to bring them to school once they have grown to compare the visible differences and work with them fresh from the soil. If no one can grow their own vegetables, try to set up a vegetable garden at the school in some corner, or just buy some different vegetables from the shops and bring them to class so learners can see the real things and describe the differences. This aspect may not have been highlighted in the curriculum, but it is important for learners to know where food comes from. And also highlight the need for schools to have vegetable gardens.

There are many different kinds of plants. If you look at different plants you can see many things that are different but also things that are the same. We know that most plants have stems, roots and leaves, and that others have flowers, seeds and fruit. If we want to compare plants, we can compare these plant structures.

You can look at the different parts of plants and compare their:

- size
- colour
- shape

Or you can ask really important questions about the plants, like:

- Does this plant flower?
- Does it lose its leaves in autumn?
- Can animals eat the plant or parts of the plant?
- Can humans eat the plant?
- Perhaps you can think of other important questions that you could ask

We are going to compare different plants using our eyes as our guides.

Look at the photos of the banana palm and the basil plant. How many differences can you see between these two plants?



A banana palm³



A basil plant.

When we compare plants, it is sometimes easier to use the different plant structures to compare the plants. We can look at the stems, for example, in the banana palm and the basil plant and compare them. The basil plant has a thin green stem while the banana palm has a thick brown woody bark covering its very thick trunk.

QUESTIONS. Describe the differences you could list between the banana palm and basil plant.

This question is specifically left open to the learners' interpretation and once everyone in the class has finished, teachers should discuss the differences and similarities the learners saw. It is hoped that some would use the stem, root and leaves to compare the two plants and teachers are encouraged to pick up on this and heap tremendous praise on these learners for thinking like Little Einsteins! This will pave the way towards the next activity.

Look at the photograph below of the edge of a river. There are main types of plants growing: the water lilies in the front and the reeds at the back. Both grow near or in water but they look completely different!

ACTIVITY: Comparing plants according to their structures



Waterlilies and reeds growing in water.



Tambo / *Phragmites vulgaris* / COMMON REED. Lu gen: Philippine Medicinal Herbs / Philippine Alternative Medicine (stuartxchange.com)



water lily roots stem leaves - Google Search

INSTRUCTIONS:

1. Work with a friend.
2. Compare the two plants (water lilies and reeds) using the plant structures.
3. When people compare different things using a set of items (like the plant structures we are using), they often use a table to write down their ideas.
4. Study the table below. Write differences and similarities between the plant structures of the water lilies and the reeds.

It is hoped that many of the learners will be familiar with either reeds or water lilies or both and that this will enrich this discussion.

Differences between water lilies and reeds

	Water lilies	Reeds
Stems	single straight thin stem, some might notice that they are not stiff but can bend with the current as visible in the one in the photo	stems are segmented, tall and woody, some might say the stems are hollow
	Cannot comment -	Cannot comment -

Roots	not visible	not visible
Leaves	round flat leaves (to float on the water)	thin narrow long leaves

TOPIC 3: WHAT PLANTS NEED TO GROW?

Conditions for growth

- How can I grow my own plants?
- If I plant seeds, what must I do to make sure that they grow?
- What does a plant actually need to stay alive and grow?

What do plants need to grow?

In this section we are going to learn specifically what makes plants grow. We will also look at growing new plants and how you can make sure that as many of your seeds as possible grow into healthy plants.

QUESTIONS

After the activity when you planted a bean seed, how did your bean grow - did it die or did it stay alive? Discuss what you think your bean plant will need to stay alive and continue growing.

Plants need:

1. sunlight, 2. water; and 3. air to grow.

1. Sunlight

- Plants need sunlight to grow
- Green plants use sunlight, water and carbon dioxide gas to make food.
- The plant can use some of this food to grow

2. Air

- Plants need air to grow. Plants use carbon dioxide in the air to make food so that they can grow.

3. Water

- Plants need water to grow. Some plants need more water than others. The amount of water a plant needs depends on the type of plant. If the plant does not get the amount of water it needs it will die. Some plants are able to grow in very dry areas, such as cacti in the desert. These plants have adapted (changed) over many, many years to be able to survive in these conditions.

- The roots of the plant absorb water from the soil. This water carries nutrients from the soil to all the parts of the plant.

Growing new plants from cuttings

We already dealt with the planting of the bean seed in the first topic.

Plants can generally be grown from seeds or cuttings.

- A cutting is made when a piece of a plant (usually the stem) is cut off and planted in new soil to start growing roots and form a new plant.

QUESTIONS

1. What three really important things do plants need to grow?
Sunlight, water, air
2. Do you remember that one of the life processes is reproduction? How do plants make new plants?
They make seeds which then germinate to grow into a new plant

TOPIC 4: STRUCTURE OF ANIMALS

In this section we are going to learn how to identify different animals. Then we will compare some animals you might already know.

Introducing this topic

This unit deals with the basic structure of animals and thus their body parts. It requires that learners compare animals based on the basic structure of animals: head, tail, body and limbs. Comparisons are based on these structural elements as well as size, shape, body covering and sense organs. Teachers are cautioned to stick to these specifications but encouraged to extend able learners by introducing such concepts as vertebrate and invertebrate, warm and cold blooded, etc.

1. Introduce this unit with a class discussion. Ask the learners to think of different animals. They can whisper their animal's name in your ear before miming the animal to the class to guess which animal they are.
2. Study the basic body structure of different animals. Use the 5 basic body parts as guidelines.

Basic structures in animals

Let's take a closer look at the body parts of animals. Just like plants, animals also have a basic structure. The basic structure of an animal is:

- head
- tail
- body
- limbs
- sense organs

Head

Most animals have a part of their body that we call the 'head'. Even the smallest animal has a part where its 'brain' is. In most animals the head has:

- a brain (or brain-like structure) - no matter how small
- sensory organs (like the eyes and ears)
- feeding parts (like the mouth and jaws)

Tail

Most animals have a tail at the back end of their body (have you ever wondered where a starfish or octopus' tail is?). A tail is often pointed but can have many other shapes as well.

Body

Different animals have different body structures and body covering.

Limbs

Most animals use their limbs to move. Animals can walk, run, climb or swim using their limbs. Some animals like mice and squirrels can use their front or upper limbs to handle objects.

Animals can have wings, webbed feet, tentacles, fins, legs, arms, flippers and long slithery bodies.

Senses

Animals have senses that help them to sense the environment around them.

Sniffer dogs help to find people who are trapped under building rubble, mudslides or snow and tell the rescue workers where the victims are. These dogs also smell drugs or bombs and alert the police.



Sniffer dogs at the airport. 15

- Eagles, buzzards, owls and other birds of prey have extremely sharp eyes as they have to see small rodents from very far away.



[Little Owl - eBird](#)

- Elephants, cats and dogs can hear sounds that human ears cannot hear.



ACTIVITY: Body plans of animals

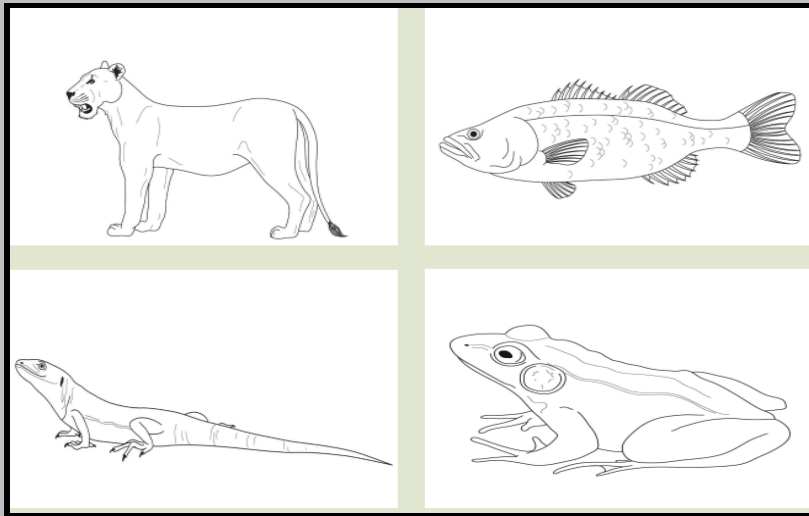
THIS REASON FOR DOING THIS ACTIVITY:

When you were learning about plants, you also learnt how to label a drawing in a scientific way. In this activity you are going to practice your labelling skills.

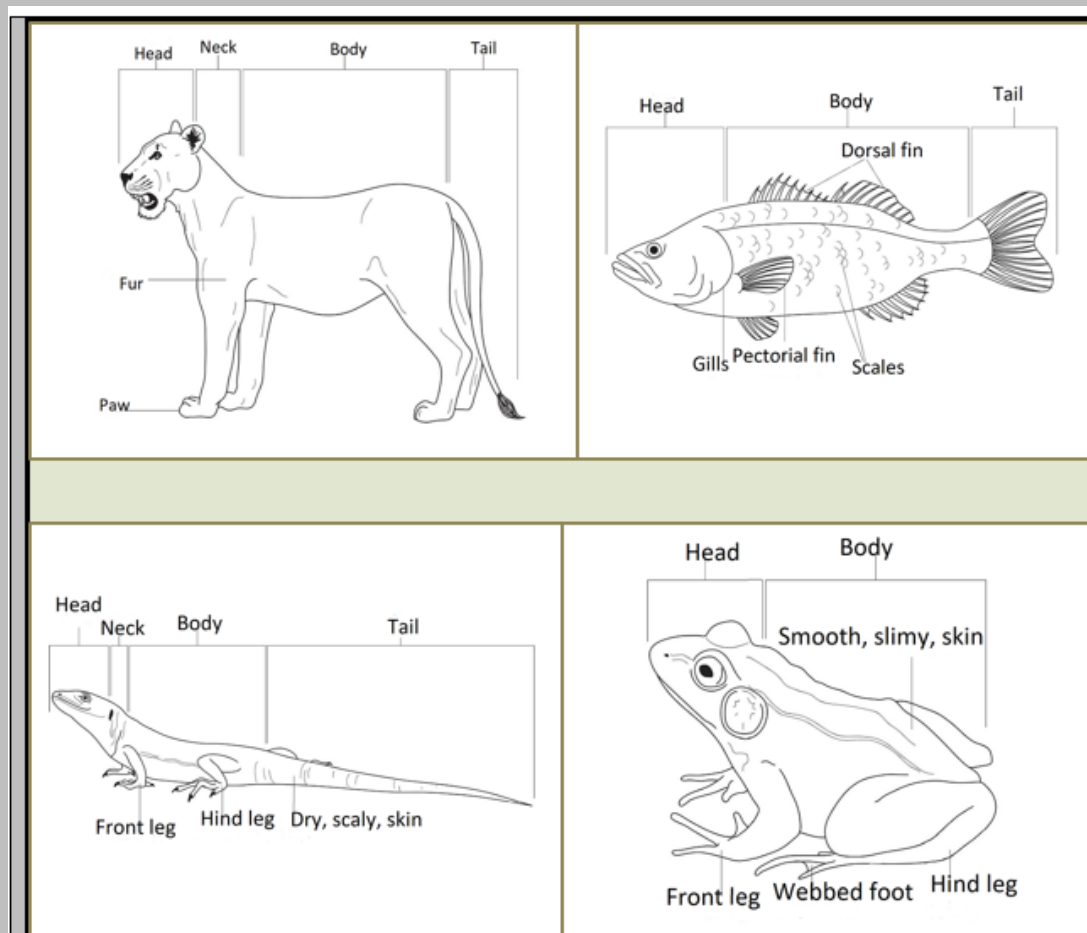
INSTRUCTIONS:

1. Study each of these drawings of different animals.

2. Use your scientific labelling skills to label each animal using the five body structures of animals



Memorandum





Grouping animals

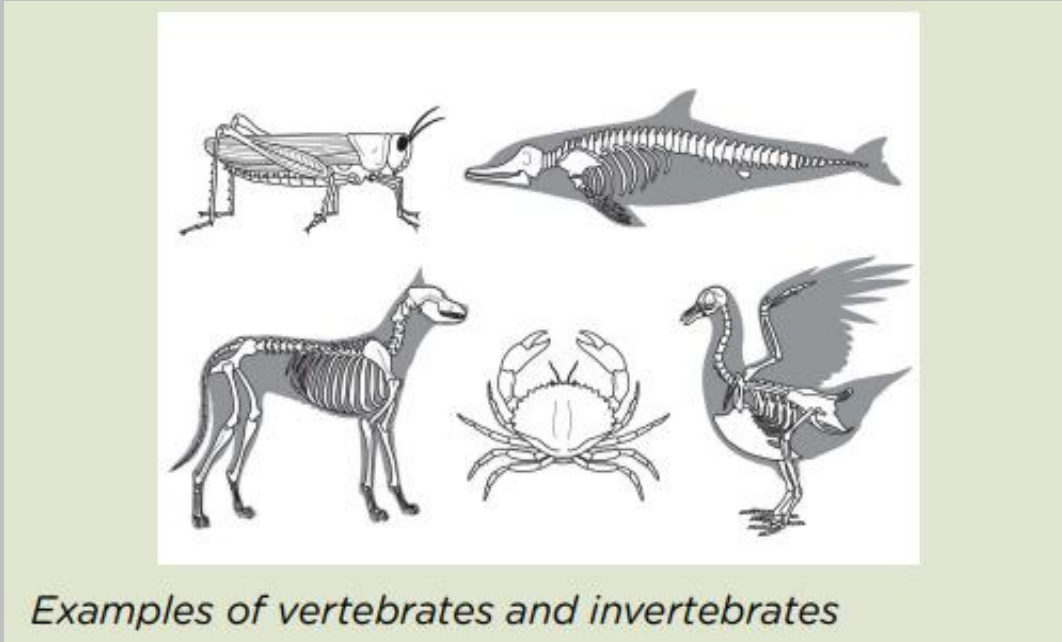
When we group similar things together, it is called classifying. When classifying animals, there are generally two main groups of animals - those who have bones inside their bodies with a backbone, and those who do not have bones inside their bodies.

- Animals **with a backbone** are classified as **vertebrates**.
- Animals **without a backbone** are classified as **invertebrate**.

ACTIVITY: Classifying animals

INSTRUCTIONS:

1. Carefully look at the following pictures and decide which animals are vertebrates and which are invertebrates.



Animals that are vertebrates	Animals that are invertebrates
dolphin, dog, seagull	Crab and grasshopper

2. Identify the visible differences between the vertebrates and invertebrates listed above.

Vertebrates	Invertebrates
Have a backbone (vertebral column)	Have no backbone (no vertebral column)
The limbs are not jointed	Jointed limbs

TERM 2: MATTER AND MATERIALS

CONTENT OVERVIEW

	Year 2	Year 3	Year 4
MATTER AND MATERIALS	<u>States of matter</u> Solids, liquids and gases - Briefly define and describe three states - Distinguish between states - Describe the properties	<u>Mixtures</u> - Briefly define concepts - Distinguish between pure substances and mixtures - Separating of mixtures (demonstrate)	<u>Acids and bases</u> - Briefly define concepts - Describe difference between acids and bases - Test various household substances with blue and red litmus paper and sort them into acids, bases and neutrals Practical activity
	<u>Metals and non-metals</u> - Define concepts - Distinguish between metals and non-metals - Properties of metals and non-metals - Uses of metals and non-metals	<u>Solutions as special mixtures</u> - Ingredients of a mixture - Explain the difference between a mixture and solution - Describe the properties of a mixture	<u>Process to purify water</u> Clean Water - Introduce and explain concepts with examples - Importance of drinking clean water - Methods of cleaning water Practical Investigation
		<u>Dissolving</u> - Explain concept - Rate of dissolving Practical investigation: Rate of dissolving	<u>Impact of weather, temperature and wind on matter and materials.</u> - Impact of weather on materials - Ways to protect materials

	<u>Processing material</u> • Briefly define concepts • Explain processes to make new materials or products • List the properties of raw materials used to make new products Practical: Demonstrate the changes of raw material	<u>The water cycle</u> • Briefly define concepts • Describe the stages of the water cycle • Draw the water cycle and describe in your own words Practical	
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CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-3	States of matter	Solids, Liquids and gases - Briefly define concepts with examples: Solid, Liquid, Gas, Evaporation, Condensation, Freezing, Melting - Demonstrate practically the 3 states in which matter could be found (solids, liquids, gases) - Distinguish between solids, liquids and gases with practical examples. - Describe the properties of solids, liquids and gases. - Describe and explore how matter can change into different states through heating and cooling using different applicable materials.
4-6	Metals and non-metals	Properties of metals - Briefly define concepts: Metals; Polymers; Ceramics - Distinguish between metals and non-metals - Name different materials in the immediate surroundings and describe their features (e.g. colour, texture etc.) - Describe the properties of different classes of metals: Appearance; Flexibility; Hardness; Shiny; Strength; Heat conduction - Describe the properties of non-metals: Dullness; Can break easy; Heat conduction; Ability to rust Uses of metals and non-metals - Research and describe how metals and non-metals are used to make products for example furniture made in the workshops at school.
7-8	Processing material	Briefly explain concepts: - Combining, Mixing, Setting, Cooking/Heating, Cooling, Drying - Explain that all materials can be processed to make new materials or products. - Explain that in Africa people have processed materials e.g. clay pots and bricks etc. - Explain that processed materials are useful because of their special properties: strong, durable, waterproof, different colours etc. - List different products and list the different raw materials it is made of. Practical: Activity

		• Demonstrate and explain processes used to change the properties of raw materials: - Mixing and setting - Mixing - Mixing and cooking - Mixing and cooling
9-10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % • Test (Pen and Paper): 60 %		

TOPIC 1: STATES OF MATTER

1.1 SOLIDS, LIQUIDS AND GASES

KEY QUESTIONS

- What are solids, liquids and gases?
- What is change of state?

Note to the teacher:

Use as much visual examples as possible, that are at your disposal

When you introduce the learners to the strand: Matter and Materials, explain to them that everything around us is made up of matter.

Introduce this section with a practical demonstration. Use examples of materials and substances to sort matter as solids, liquids and gases.

You will need the following materials: wood, stone, plastic, a glass of water, another container to pour the water in, juice, tea, air (in two or three different shaped balloons or tyres), cooking oil, cooking gas, a boiling kettle etc. Introduce the learners to the examples. Let them observe, feel, smell and touch the examples. Let the learners engage in the activity to identify solids, liquids and gases

Everything around us is made up of matter. Matter takes up space and has mass, this means we can weigh matter. Solids, liquids and gases in the universe are matter.

Solid: Substance that has a definite shape and volume. Solids could be hard, soft, big or small. Some examples are stone, wood, towel and sand.

Liquid: Substance that flows freely. Liquids have definite volume but no definite shape, they take the container they are in. Some examples are water, paraffin, oil, fruit juice, petrol or methylated spirits.

Gas: Substance that expands to fill the space in it. No fixed shape and no fixed volume. Some examples are helium, oxygen and carbon dioxide.

ACTIVITY: Exploring the properties of solids

Materials:

- a stone
- cloth
- paper
- wood

Questions

Use the questions below to investigate each solid.

Answer the following based on each solid

ACTIVITY

USE THE TABLE BELOW TO FILL IN SOME OF YOUR ANSWERS ABOUT EACH OF THE OBJECTS

OBJECT	QUESTIONS	OBSERVATION
Stone	1. Does it feel hard or soft? 2. Does it break easily? 3. Does it change its shape when you put it in something else? 4. How would you describe the shape?	
Cloth	1. Does it feel hard or soft? 2. Does it break easily? 3. Does it change its shape when you put it in something else? 4. How would you describe the shape?	
Paper	1. Does it feel hard or soft? 2. Does it break easily? 3. Does it change its shape when you put it in something else? 4. How would you describe the shape?	
Wood	1. Does it feel hard or soft? 2. Does it break easily? 3. Does it change its shape when you put it in something else? 4. How would you describe the shape?	

QUESTIONS

- Write down common properties for all the solids you investigated?
 - definite shape, hard, soft
- List some other solid objects in your classroom. Give at least **two** examples.
 - Wood, door, ruler, pencil, rubber, desk, etc.

We have learned that a substance in a solid form will have a fixed shape and takes up a definite space. **Let's now look at liquids**

ACTIVITY: Exploring the properties of liquids

MATERIALS:

- water
- oil
- fruit juice
- 3 small pieces of cloth
- 3 containers for each of the 5 liquids
- 3 other clean and empty containers, such as a glass, cool drink bottle or tin
- 3 saucers

INSTRUCTIONS:

Study the different liquids on the table and answer the following questions.

Write your answers in the table that follows.

DO NOT TASTE THE LIQUID!

WASH YOUR HANDS AFTER HANDLING THE LIQUID

OBJECT	QUESTION	OBSERVATION
Water	1. Can you soak the liquid up with a cloth? 2. Can the liquid flow or spread out on a saucer? 3. How will you describe the shape of the liquid, is it fixed, does it take the shape of the container?	
Oil	1. Can you soak the liquid up with a cloth? 2. Can the liquid flow or spread out on a saucer? 3. How will you describe the shape of the liquid, is it fixed, does it take the shape of the container?	
Juice	1. Can you soak the liquid up with a cloth? 2. Can the liquid flow or spread out on a saucer? 3. How will you describe the shape of the liquid, is it fixed, does it take the shape of the container?	

QUESTIONS

- Write down the safety rules for this investigation.
 - Don't taste any unknown liquid, wash your hands after handling unknown liquid.
- Write down the common properties for all the liquids investigated.
 - Liquids flow and can be poured, has no fixed shape and they take the shape of the container.

After doing this activity where we investigated the properties of liquids, we can say that a liquid:

- can flow,
- has no fixed shape; and
- takes the shape of the container that it is in.

Now that we know the properties of solids and liquids, let us investigate gases.

Gas

When we breathe, we are taking in and giving out gases. But we cannot see the gas. Most gases are invisible and there are few that are visible. We use gas to cook, and hospitals also use gas for patients to breathe.

The air you breathe in has oxygen gas. The air you breathe out has more carbon dioxide gas.

ACTIVITY: Learning about gases from pictures

INSTRUCTIONS:

- Study the pictures below. Each of the pictures shows a different property of a gas.
- The properties are listed in the first column of the activity below. Decide which picture is showing the property and give it a tick. Make a tick under the right picture for each property.
- The first one has been done for you.

Property illustrated			
A gas moves without something		✓	

that pushes it - it diffuses through the air.			
A gas has no definite shape and fills the container it is in.	✓		
A gas can be pressed to fill a smaller space.			✓

In summary, gas

- takes up all the space available.
- has no definite shape.
- can flow.

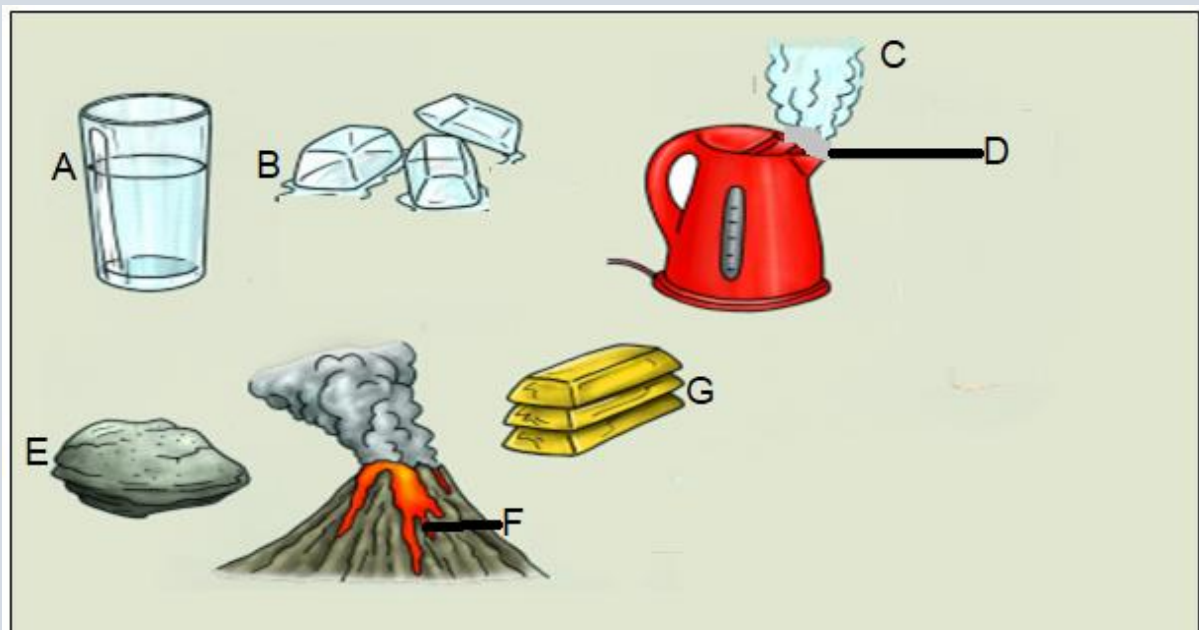
Now that we have learnt about solids, liquids and gases, let's see if we can identify solids, liquids and gases

ACTIVITY: Identifying solids, liquids and gases

INSTRUCTIONS:

Look at the pictures of the different substances below and decide if they are solids, liquids or gases.

Use the table below place a tick in the right column.



SUBSTANCE	SOLID	LIQUID	GAS
A. water in the glass		✓	
B. Ice blocks	✓		
C. Condensed water (cloud)		✓	
D. Steam			✓
E. Rock	✓		
F. Lava from volcano		✓	
G. Gold bars	✓		

Comparing solids, liquids and gases

SOLIDS	LIQUIDS	GASES
Have a definite shape	Have no definite shape	Have no definite shape
Takes up a definite space	Takes up a definite space	Takes up all the space available
Do not flow	Can flow	Can flow
		
		<i>balloons filled with helium gas</i>

<https://cdn.shopify.com>

<https://www.bing.com>

1.2 CHANGE OF STATE

We learnt about the states of matter. These were solids, liquids, and gases. A substance can change from one state to another. For example, a solid can change into a liquid, water can be a liquid in your glass or turn into ice in the freezer. Ice is a solid. But what makes these substances change from one state to another?

Freezing and melting

Let's use water as an example.

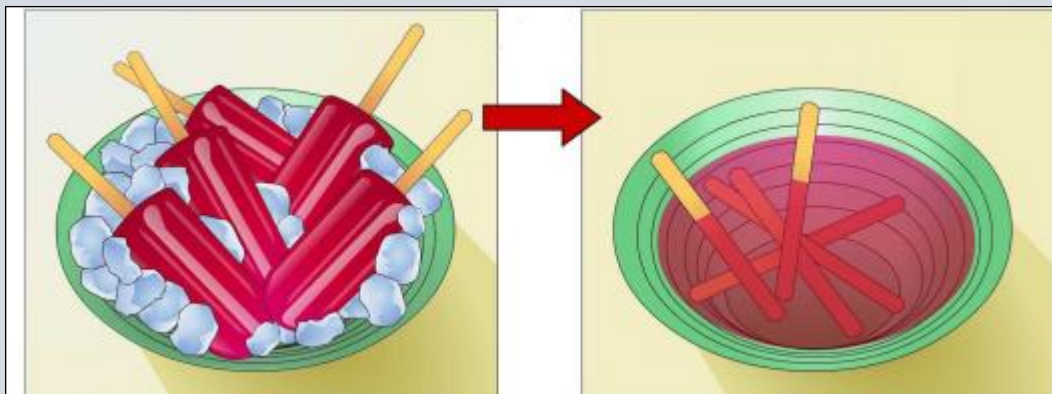
QUESTIONS

1. If you place tap water into an ice tray and put this in the freezer, what will happen to the water?
 - The water freezes
2. If you now take ice cubes and place them in the sun, what happens to the ice cubes?
 - The ice melt.

If we place the water in a freezer, it freezes. If we place the ice cubes outside the freezer, they melt. This is because the state of matter can be changed from one to another by adding or removing heat.

ACTIVITY

Look at the picture of frozen ice lollies that were left in the sun and answer the questions



QUESTIONS

1. When the ice lollies are frozen, are they a solid, liquid or a gas?
 - They are solid
2. What happened to the ice and the ice lollies while they were left in the sun?
 - Heat from the Sun causes a state change and the ice lollies changed (melts) from a solid to liquid.
3. What do we call this process?
 - Melting.
4. Can the process of melting be reversed?
 - Yes, you can freeze the liquid so that it becomes frozen again.

The liquid ice lollies were put into a freezer. They changed from liquid to solid due to reduction of temperature. This is called **freezing/solidifying**.

The ice lollies were frozen solid. When they were placed in the Sun, they started to turn into liquid. The heat from the Sun caused change of state from solid to liquid. This change from solid to liquid is called **melting**.

Note to the teacher: Do the following activity as a demonstration.

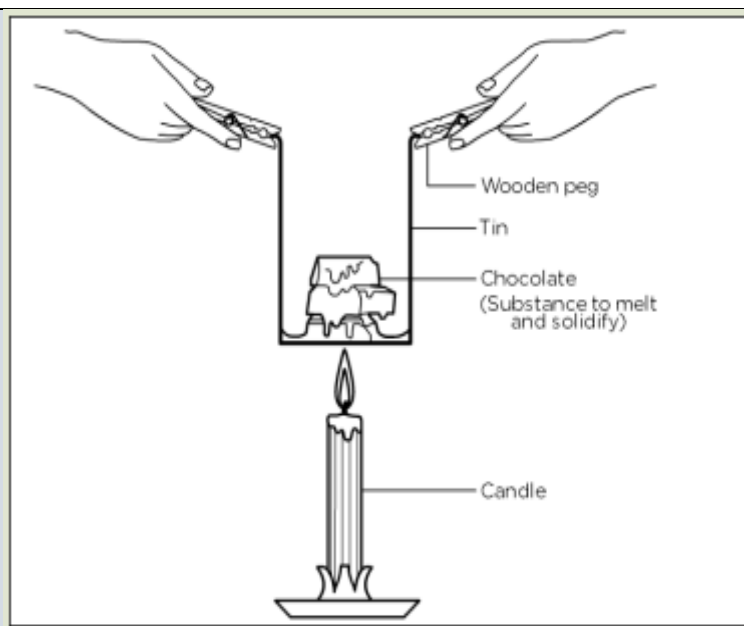
ACTIVITY: Melting and solidifying substances.

MATERIALS:

- butter, fat or margarine
- chocolate or wax
- ice blocks or ice cream
- 3 containers which will not melt (they can be empty tins)
- 6 wooden pegs
- a candle
- matches

INSTRUCTIONS

1. Plan how you are going to melt and solidify the substances.
2. Look at the diagram below which shows how you can do this.
3. Test each substance that you have by placing it in the tin and holding it over the candle.
4. Then remove the tin from the candle and leave it on the side to cool.
5. Observe what happens to the state of each substance and write down your observations in the table below.



Setup for the demonstration

Substance	Observation before heating	Observation after heating	Observation after cooling
butter/margarine	solid	liquid	solid
chocolate/candle wax	solid	liquid	solid
Ice blocks/Ice cream	solid	liquid	liquid

QUESTIONS

1. What happened when the solids were heated by the candle?
 - They melted
2. What happened to the substances when they cooled down again?
 - They solidified except for ice
3. What do we call the process when a liquid changes to a solid?
 - Solidifying / Freezing

Condensation

Note to the teacher

- The following activity is to be done as a demonstration in the front of the class.

- Invite learners up to your desk in small groups to see the water boiling and to observe the cloud of water as it hits the mirror and condenses.

ACTIVITY: Heating and cooling to cause a change of state

MATERIALS

- kettle
- liquid water
- glass or mirror
- gloves or towel

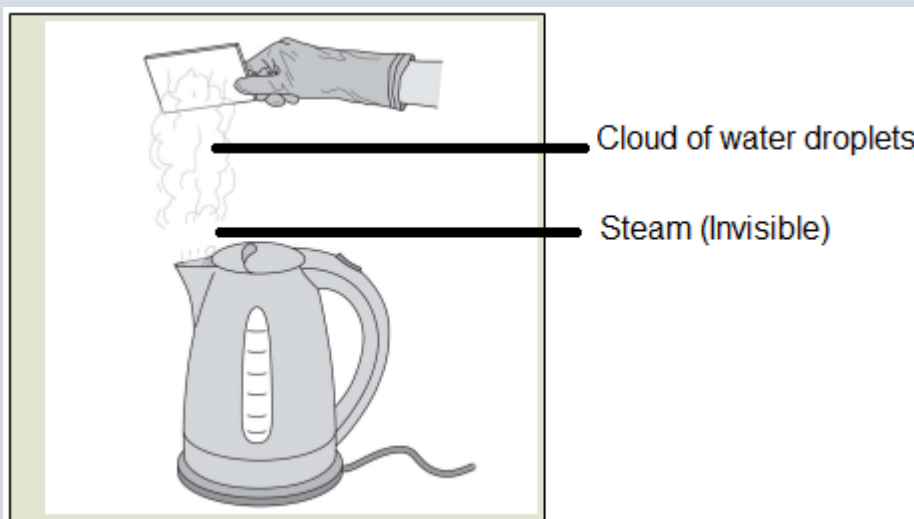
INSTRUCTIONS

Boil the water in the kettle.

Put a glass or mirror 30 cm above the boiling kettle (you need to wear gloves made of thick material or use a towel to avoid burning your skin)

Be careful not to introduce the misconception that water needs to boil in order to evaporate. Even cold water can evaporate at room temperature. The steam is not visible coming out of the kettle. The steam is actually just at the spout when it first comes out and is extremely hot. The "cloud" that you see forming is actually when the steam has already started to cool and condense and form water, but in tiny droplets which are visible. This "cloud" is not a gas but actually tiny droplets of water in the air. The use of the mirror is to help speed up the process of condensation and show what is happening.

This activity shows boiling and condensation, not necessarily evaporation.



1. What do we call the process when the steam changes into a cloud of water droplets?

Condensation (change gas to liquid)

The steam that comes out of the kettle is extremely hot and you cannot see it. The steam quickly cools and forms tiny droplets in the air. These tiny droplets are visible and form the "cloud" that you see. When these tiny droplets hit the mirror they cool more and form the bigger droplets which you see forming on the mirror. The change of state from the gas (water vapour) state to the liquid state is called **condensation**.

When you leave a glass filled with cold water on the table, small droplets form on the outside. This is because there is water vapour in the air which cools down when it is near the cold glass. The water vapour in the air around the glass condenses and changes from gas to a liquid and forms the tiny droplets you can see.



Water droplets on the outside of a cold glass

Evaporation

Evaporation takes place only at the surface of a liquid at any temperature. Evaporation is the process of a liquid changing into gas.



We hang wet clothes out to dry. They dry as the water evaporates.

Here is a summary of the different state changes:

Change of state	We call the process
Solid to liquid	Melting
Liquid to gas	Evaporation
Gas to liquid	Condensation
Liquid to solid	Freezing and solidifying

REVISION

1. What are the three states of matter?
 - Solid, Liquid, gas
2. Describe what happens to solid ice when it is heated.
 - The temperature rises causing the ice (solid) to melt
3. Below are the definitions of each of the three states of matter. They are placed in the wrong order. Match the right letter to the number and complete the last column.

Definition	State	Answer
3.1. Has a definite shape and takes up a definite space on the surface.	Liquid	B
3.2. Has no definite shape and spreads in the space.	Solid	C
3.3. Has no definite shape and takes the shape of a container.	Gas	A

4. What will happen to the water in a saucer if we leave it uncovered for four hours?
 - It will evaporate.
5. Name 2 properties of solid.
 - Fixed shape, dry, hard, fixed volume
6. A block of ice, is left outside the freezer. Discuss what changes you would observe in the object after three hours.
 - The block of ice will melt and become a liquid very quickly.
7. What is the reverse of freezing?

- Melting

Summary

- Matter is everything around us.
- Matter can be solid, liquid, or gas.
- Solids are matter that has a fixed shape.
- Liquids flows, can be poured, takes the shape of the container.
- Gases are mostly invisible and takes the shape of the container and spreads out / flows in space.
- A change of state is brought about by heating and cooling matter.

TOPIC 2: METALS AND NON-METALS

KEY QUESTIONS

- How can we tell if something is made of a metal or a non-metal?
- How do we decide what material to use when we want to make or build something?

Note to the teacher:

Make use of visual and practical examples in their surrounding area and workshops at school. You could draw their attention to metallic and non-metallic items in the classroom, or in pictures, as an introduction to the topic and related activities.

In this section we investigate and contrast the properties of metals and non-metals. Metals and non-metals are two different classes of materials. Each class has its own unique properties.

2.1. Metals

Metals are shiny solids that can conduct heat or electricity and can be formed into sheets or wire. Some examples of metals are copper, iron, silver.

Properties of metals

- Metals are usually shiny.
- Most metals are hard and they feel heavy.

- Metals conduct electricity and heat well.
- Metals are malleable (they can be shaped into flat sheets) and
- They are ductile (this means they can be made into thin wire)

2.2. Non-metals

These are any materials that are not metals. Some examples are wood, chalk, plastic, etc.

Properties of non-metals

Non-metals are materials that do not have the same properties as metals.

- Non-metals are not shiny, but tend to be dull.
- Many non-metals are not bendy (flexible) but brittle. This means that they will break when we try to bend them with enough force.
- Non-metals do not conduct electricity or heat well

ACTIVITY: What are the properties of metals and non-metals?

AIM: To describe the properties of metals and non-metals

MATERIALS:

- coins
- metal spoon
- metal nail or screw
- steel wool
- cardboard
- cotton wool
- plastic spoon
- piece of chalk

METHOD:

1. Few objects have been placed in front of you. Notice all the different shapes. Write the name of each object in the table below.
2. Hold each object in your hand. Is it rough or smooth?
3. Look at each object carefully. Is it shiny or dull? Can you describe its colour?
4. Drop each object on the floor, or tap it. What sound does it make?
5. Dip the metal spoon and plastic spoon in a cup filled with hot water for 5 minutes. Feel the handle of the spoons. What do you feel? (Is it hot/cold)

OBSERVATIONS

Fill in the observations from your investigation of different metals and non-metals below:

Object	Shiny/dull	Malleable	Hardness	Heat conduction	Metal or non-metal
Coins					
metal spoon					
metal nail or screw					
steel wool					
Cardboard					
cotton wool					
plastic spoon					
piece of chalk					

CONCLUSION

What have you learned from investigating the properties of metals?

Distinguish between metals and non-metals

What have we learnt about the properties of metals and non-metals?

Now we are ready to compare the properties of metals and non-metals.

Metals	Non-metals
Good conductor of heat	Poor conductor of heat
Good conductor of electricity	Poor conductor of electricity, except for graphite
Malleable (can be hammered)	Not malleable (cannot be hammered)
Ductile (this means they can bend without breaking)	Not ductile (this means they cannot bend without breaking); they are brittle
Shiny or silvery (lustrous)	Dull

ACTIVITY

Research and describe how metals and non-metals are used to make products for example furniture made in the workshops at school.

2.3. USES OF METALS AND NON-METALS

Note to the teacher:

This section focuses on ways in which metals and non-metals are used. Draw learners' attention to metallic objects in and around the classroom. They could each bring one picture of a metallic object and non-metallic objects to class and these could then be sorted into categories, for instance 'transport', 'the kitchen', 'industry', etc. Learners could generate the categories themselves, and then be encouraged to think about the reasons why metal or non-metal is used for each particular purpose.

Metals and non-metals have many uses. We use them every day, sometimes even without knowing!

Here are some of the uses of metals

- Copper is used for making electrical wires, cooking utensils
- Iron or steel is used to construct the house and bridges
- Zinc is used to protect the iron from rusting
- Gold and silver are used for making jewellery

The pictures highlight some of the uses of metals



Aeroplanes are made from strong, durable sheets of metal.



A bridge made of metal.

Here are some of the uses of non-metals

- Oxygen is used for breathing
- Chlorine is used for purifying water
- Nitrogen is used in fertilizers
- Graphite is used as a good conductor of electricity

The pictures highlight some of the uses of non-metals

Summary

- Every type of matter has its own set of properties
- Metals and non-metals have different properties.
- Materials are useful because of their properties.
- Metals and non-metals have many uses.

REVISION

1. List 3 properties of metals.

- Metals are solid, strong, malleable, ductile, and lustrous.
In addition:
- Some metals are magnetic;
- metals conduct heat and electricity;
- and some metals corrode (for instance iron rusts).

2. Why do some pots and pans have handles made of plastic or wood?

- Metal conducts heat but plastic and wood do not. That means the handles will stay cool even when the pot is too hot to touch

TOPIC 3: PROCESSING MATERIALS

KEY QUESTIONS

- How can we make new materials?
- How does the amount of material we start with affect the amount of new material we can make?

This topic deals with the many different ways in which materials can be processed. The important messages are that new materials will have new properties, and that processing always has a purpose.

There are many ways to process existing materials into new materials. There are also many reasons why we would need to process materials into new materials. When we bake a cake, we are processing flour, eggs and other ingredients (that may not taste very nice on their own) into cake which tastes really good! We process materials to make them stronger, or more durable, or waterproof, or even just to make them look more beautiful or interesting. New materials that form after mixing different materials are sometimes called mixtures.

3.1 DEFINITION OF CONCEPTS

- **Combining**
Mixing different materials to make new products
- **Mixing**
Combining of two or more materials to achieve a desired level of uniformity
- **Cooking**
Skill of preparing food by combining, mixing and heating ingredients
- **Heating**
Conversion of energy to heat
- **Cooling**
Removal of heat
- **Drying**
Removal of water by evaporation from a solid

Materials can be processed to make new materials or products.

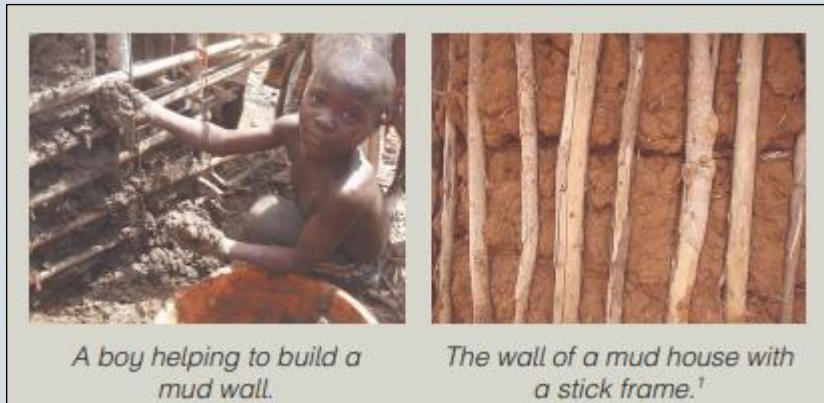
When we combine materials, new materials are made. The properties of the new material are often different from the properties of the material we started with. There are many ways to process existing materials and many reasons why we would need to process materials into new materials. This might include materials that can mix and set, such as plaster of Paris and water, baking a cake or using jelly powder.

3.2. PROCESSED MATERIALS MADE BY PEOPLE IN AFRICA**Traditional processing**

- In the olden days only natural materials were available and people found many clever ways to make these materials more useful.
- They also had ways of reinforcing the mud used for making traditional huts.
- They also knew exactly which reeds would make the best mats to cover their walls, and how to build the best houses for their climate and lifestyle.
- Some of these traditional ways of processing materials are still used today.
- We will learn about a different kind of traditional home, which is still seen today.
- Some of the traditional homes in Africa are made of clay or mud.
- Mud (mixture of soil and water). When it is reinforced it can make a strong and durable building material, which can be used to build a house. If it is well built, the house will last for many years.

ACTIVITY: Making a wall of mud house

Study the photos. Then answer the questions.



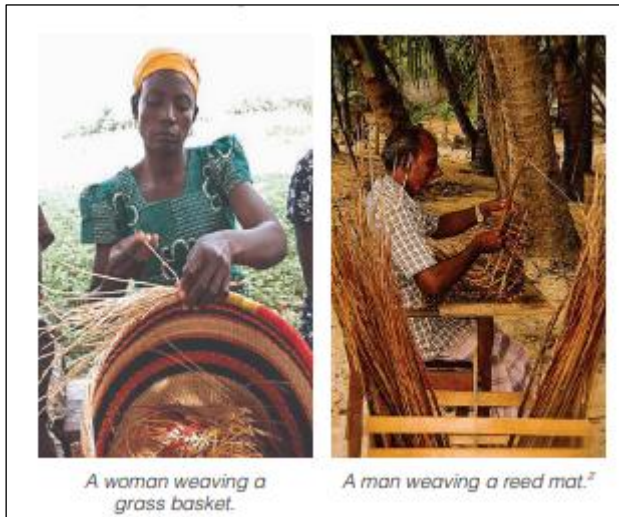
1. What material is the house in the pictures made of?
 - Mud, sticks
2. Look at the close-up photo of a wall, to see how he strengthened the wall of his house. Name the method used to strengthen the walls.
 - He used sticks to build a framework for the house
3. List three reasons why we process materials
 - We process materials to make them stronger,
 - more durable,
 - waterproof,
 - fire-resistant,
 - more beautiful (any sensible reason is correct).
4. Below are two pictures. Describe the properties of the materials in both pictures and what processes took place to get from Picture 1 to Picture 2.



In **Picture 1** the clay is wet and soft. It has been mixed with some water so it can be moulded.

In **Picture 2** the clay is hard and dry. But it is also brittle as if you drop it on the floor it will smash. To get from the clay in Picture 1 to the pot in Picture 2, the clay was moulded into a specific shape and then left to dry. After drying it was fired in a kiln to bake it and make it set. It was also painted at the end.

In Africa, many people make objects by using plant products, called plant fibres. The people weave and stitch the plant fibres together to make different objects, such as reed mats, baskets, or even thatch to make a roof for a house. This is also a type of traditional processing



ACTIVITY: Identifying objects made from plant material

INSTRUCTIONS

1. Each of the following photos shows an object made from plant material.
2. Identify what it is and how the people are using these objects.

OBJECT MADE FROM PLANT MATERIAL	DESCRIPTION
	This is a thatched roof made from grass used as a shelter
	A hat used to protect their face from the sun.
	Houses are made from reed mats.

3.2 METHODS FOR PROCESSING MATERIALS

The main focus will be on **combining materials** and **how they can be processed to make new materials and products**. This might include materials that can mix and set, such as plaster of Paris and water. We will also look at baking a cake or using jelly powder in the classroom to demonstrate this process.

Combining materials

Process: To process material means to combine or mix it with other materials and/or to change it into something new by subjecting it to one or more processes (heating, cooling, firing, melting, etc.)

Let's look at more ways to combine and process materials.

Note to the teacher:

This is a very quick and easy activity to demonstrate the concept of the finished product having different properties to the starting materials. Place the flour and water in separate bowls and allow learners to put their hands in to describe the properties. It

might get a bit messy for learners to each do their own mixing, so you could do it as a demonstration at the front of the class.

ACTIVITY: Making glue (Mixing)

Often, when we mix materials together, the properties of the new material or product is different from the properties of the materials we started with.

Let's make some sticky glue to find out about this!

MATERIALS:

- flour
- water
- 2 bowls for the flour and water
- a bowl for mixing
- pieces of paper

INSTRUCTIONS

1. Firstly, discuss the properties of the flour and water before mixing them together.
2. Now experiment with mixing different quantities of water and flour together to make a sticky paste.
3. See if you can stick pieces of paper together using the glue that you have made!
4. Describe the properties of the paste you have made.
5. Is the final product after processing a totally new product?

QUESTIONS

1. What did the flour feel like before it was mixed with the water?
Dry, powdery, soft
2. How would you describe the properties of plain water?
Wet, liquid, can pour, fills the container it is placed in, etc.
3. After mixing the flour and water together, what are the properties of the paste that you end up with?
Sticky, wet, more solid than water, etc.
4. What state of matter is the flour before mixing?
Flour is a solid
5. What state of matter is the water before mixing?
water is a liquid.

Did you get any of the paste you made in this activity on your fingers? Perhaps it started to dry and become hard? When we combine some materials together we have to let them set to get the desired product.

Note to the teacher:

Jelly needs to set over some time in a cool place, and this should be kept in mind when starting this activity. It would be best if the educator handled the hot water, rather than letting the learners do this. Some learners could bring jelly to school and other could bring fruit to slice into the jelly before it sets. The jelly could also be set in small yogurt containers or ice cube trays so that each learner can enjoy it the next day.

ACTIVITY: Making Jelly (Mixing, cooling and setting)

Have you ever tasted jelly? Jelly comes in many different colours and flavours. Which colour is your favourite? To make jelly, we must dissolve jelly powder in hot water. When the solution of jelly powder in water cools down, something very special happens: The solution sets, and turns into a delightfully wobbly, sweet treat! The jelly powder has been processed into something new! That is what the next activity is all about.

MATERIALS:

- a packet of jelly powder
- a bowl
- a measuring cup
- hot and cold water
- spoon for mixing

INSTRUCTIONS:

1. Read the instructions on the packet of jelly.
2. Pour the jelly powder onto the bowl.
3. Look carefully at the dry jelly powder. What does it look like?
4. Write your findings in the table below.
5. Follow the instructions on the packet to make the jelly.
6. Cool the jelly until it sets.
7. Describe the properties of the prepared jelly in the same way that you did for the starting materials.
8. Is the final product after processing a totally new product?

Here are some words that you may find useful. You may also use your own words.

liquid, clear, powdery, sweet, sticky, transparent, wobbly, solid, gelatinous (jelly-like), slippery

TABLE OF OBSERVATION

Properties	Dry jelly powder (before mixing)	Water (before mixing)	Prepared jelly (after it has set)
What does it look like?			
What does it feel like?			

QUESTIONS

1. What materials did you start with?
 - Jelly powder, water, (fruit is optional).
2. What happened to the jelly powder when you mixed it with water?
 - Learners dependent answer, they may write something like: The jelly powder dissolved in water and became a solution, etc.
3. Why is the water a different colour?
 - The Jelly powder mixed with the water and this changed the colour of the water
4. How did the jelly mixture change when it cooled down?
 - The jelly mixture was liquid at first but it became stiff when it cooled.

Mixing & Cooking

Mixing and cooking food is also a form of processing.

- Mixing is combining together two or more ingredients.
- Cooking involves heat. It includes methods such as boiling, baking, steaming, roasting and frying

Look at the pictures below:





source: <https://www.allrecipes.com>

Now that we know that materials can be processed, let us now explore the properties and uses of processed materials

3.4 PROCESSED MATERIALS

Special Properties of processed materials

Processed materials are strong, durable (long lasting), waterproof, different colours (interesting and attractive), fire and heat resistant and cost effective.

Plaster of Paris

- Made from gypsum
- White powder is mixed with water to form a paste that can be moulded into various shapes. When it dries it becomes very hard.
- It is water proof and fire resistant
- Used as plaster cast, dental moulds, art works, fill cracks

Concrete

- Made from cement, water, gravel and sand
- Is an important material used in building many things: roads, pillars, pipes, bridges, etc.
- It is strong and hard
- It is durable
- It is fire resistant
- It is waterproof
- It can easily be moulded into any shape

Ceramics and glass

- Ceramics items include casserole dishes, plates and mugs, etc.
- Glass include glass cups, jugs and windows, etc.
- Very useful because they are water proof
- Can withstand high temperatures (heat resistant)
- We can use them for cooking

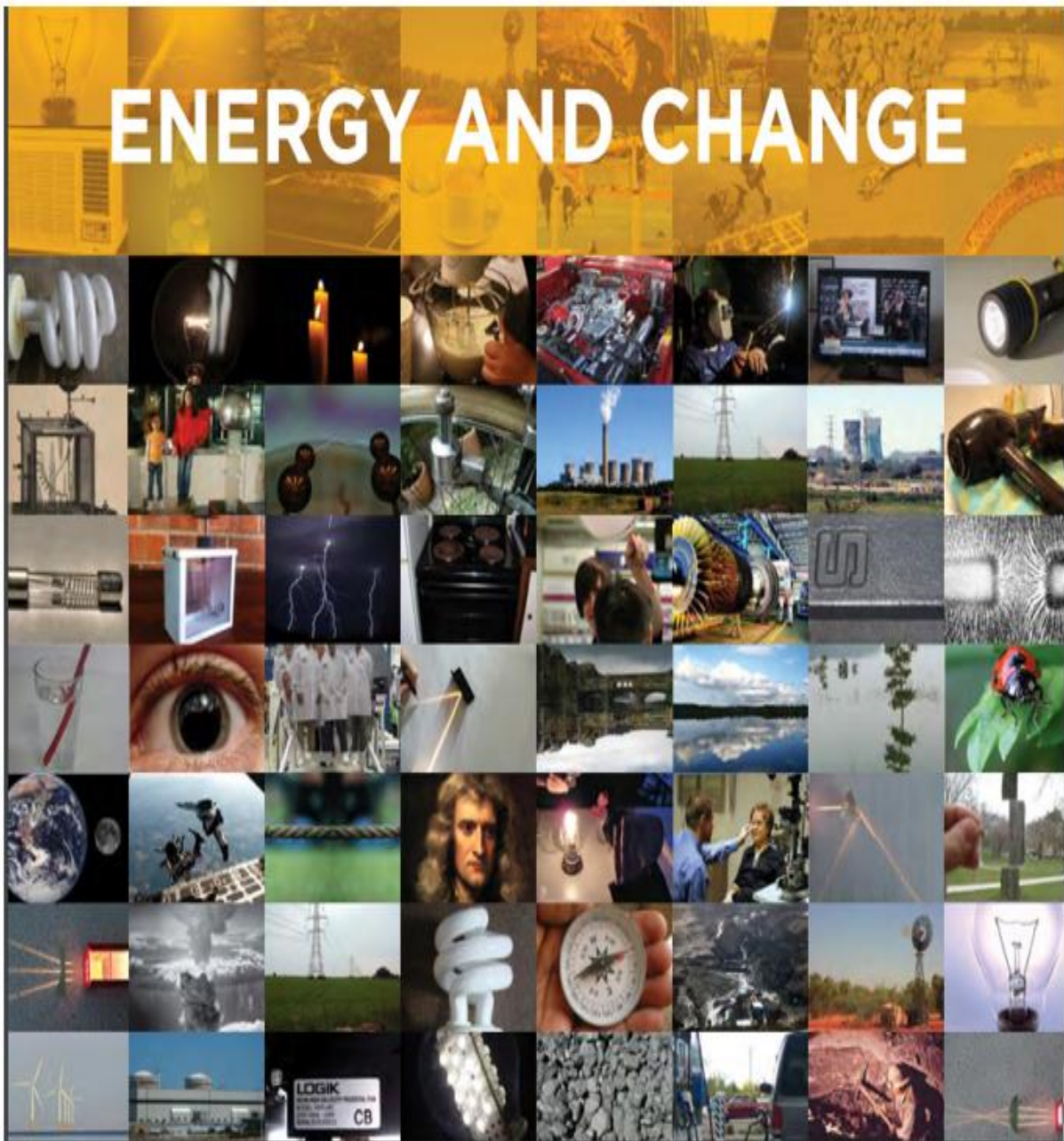
Fabric

- Made of fibres that are either natural or man made
- Natural fibres come from animals or plants e.g. Cotton from plant, wool from sheep, silk from silkworm
- Manmade fibres are made from other substances e.g. Nylon and polyester
- Wool jersey trap the warm air from our bodies which helps to keep us warm
- Cotton is cooler to wear.
- Towelling fabric is used to make towels and dish clothes.

Summary

- Materials can be processed in many different ways to make new materials or products.
- When we process materials, the new materials may have different properties.
- The purpose of most processing methods is to make materials more useful.
- In Africa, people have processed materials for many years, for example to make clay pots and woven products.

TERM 3



Term 3: ENERGY AND CHANGE

CONTENT OVERVIEW TERM

STRAND	YEAR 2	YEAR 3	YEAR 4
Energy and Change	<u>Energy</u> - Explain concepts with examples - Different forms/types of energy	<u>Energy and Temperature</u> - Explain the concepts with examples - Describe how to read and use a thermometer - Describe how heating and cooling can change materials	<u>Electric circuits</u> - Define concepts - Explain when a circuit is complete - Conductors and Insulators
	<u>Energy Sources</u> - Explain concepts with examples - Categorise sources of energy into Renewable and Non-Renewable sources - Name the different energy sources	<u>Stored energy in fuels</u> - Explain concepts with examples - Explain how energy is stored in food - Discuss the Renewable and Non-renewable sources	<u>Systems to solve problems</u> - Identify different circuits - Make simple electrical systems
	<u>Energy Transfer</u> - Explain where the energy from food comes from - Describe how energy is transferred in a food chain - Identify different types of energy transfer	<u>Burning Fuels</u> - Investigate, compare & describe how different fuels burn: Example candle <u>Safety with fire</u> - Identify sources - Identify safety rules when using fuels	<u>Mains electricity</u> - Describe concepts with examples - Compare the amount of energy released by different energy sources
	<u>Sound and Noise pollution</u> Explain the concept sound with different examples	<u>Energy and Electricity</u> - Briefly define concepts - List the different types of batteries	<u>Saving Cost of Generating Electricity</u> Explain concepts with examples

	• Explain sound as noise pollution and the impact on the environment	and its uses	• Describe energy sources and usage in South Africa • Compare different electrical appliances
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ENERGY AND CHANGE CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-5	Energy and Energy Transfer	Energy from the Sun • Explain concepts with examples - Energy - Energy from the sun Energy around us Practical: Activity • Identify and demonstrate different types of energy transfer (examples) - electrical energy to: - heat energy - light energy - sound energy - movement energy - food energy to movement energy (Make use of a variety of examples) Doing this work practical is very effective.
6-7	Energy Sources	Energy Sources • Explain the following concepts with examples: - Energy sources (main source of energy- Sun, food, - Food as a source of energy - Food chain • Describe the organization of a simple food chain. - Energy from the Sun. - Producers (plants) - Users (Herbivore, Carnivores, Omnivores) - Decomposers - Renewable sources of energy e.g. Sun, wind and water. - Non-renewable sources of energy e.g. coal, crude oil, natural gas, fuels. • Discuss the classifications of both Renewable- and Non-

		Renewable sources of energy:
8	Sound and noise pollution	Noise pollution • Briefly explain the following concepts with examples: Sound; Noise pollution • Explain what noise pollution is and give practical examples in their environment. • Describe the danger of: - Loud sound - Sound to hearing
9-10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % • Test (Pen and Paper): 50 %		

Topic 1: Energy

Key questions

- What is energy?
- What do we need energy for?
- Where does energy come from?

What is energy?

Energy is the ability to do work. We get energy from various sources



What do we need energy for?

Note to the teacher: The activity below is for learners to have some fun playing a game so that the learners are tired afterwards. You can then introduce the idea that they used energy to run around.

We need energy to carry out all our life processes. Without energy, there would be no life.

ACTIVITY: Energy is the ability to do work

Let us have some fun playing a game!



Learner playing hide-and-seek

INSTRUCTIONS:

1. Divide learners into manageable groups and go outside for 10 minutes to play "Hide and Seek", or any game that will make them move.
2. These are the rules of the game:
 - a) One person needs to be the seeker.
 - b) The seeker needs to find a home - a tree or the wall will work well.
 - c) The seeker closes their eyes and counts to 20.
 - d) Everyone else must hide. They are the hiders.
 - e) When the seeker has counted to 20, (s)he must find the others and tag them by touching.
 - f) If the hiders can get to the seekers' "home" without him or her seeing them, they can block themselves and cannot be tagged by the seeker.
 - g) If the seeker finds a hider s/he runs to their "home", touches it and says "1, 2, 3, Tag Emma!" (or whoever they saw).
 - h) If there is time left over, you can choose a new seeker and play again.
3. Return to class after 10 minutes and talk to in your group about how you feel.
4. Each group must choose a leader who will report back to the class.
5. Write down the words that are being used to describe how you are feeling after the game.

You will feel tired from the game. This means that energy is being used to do work.

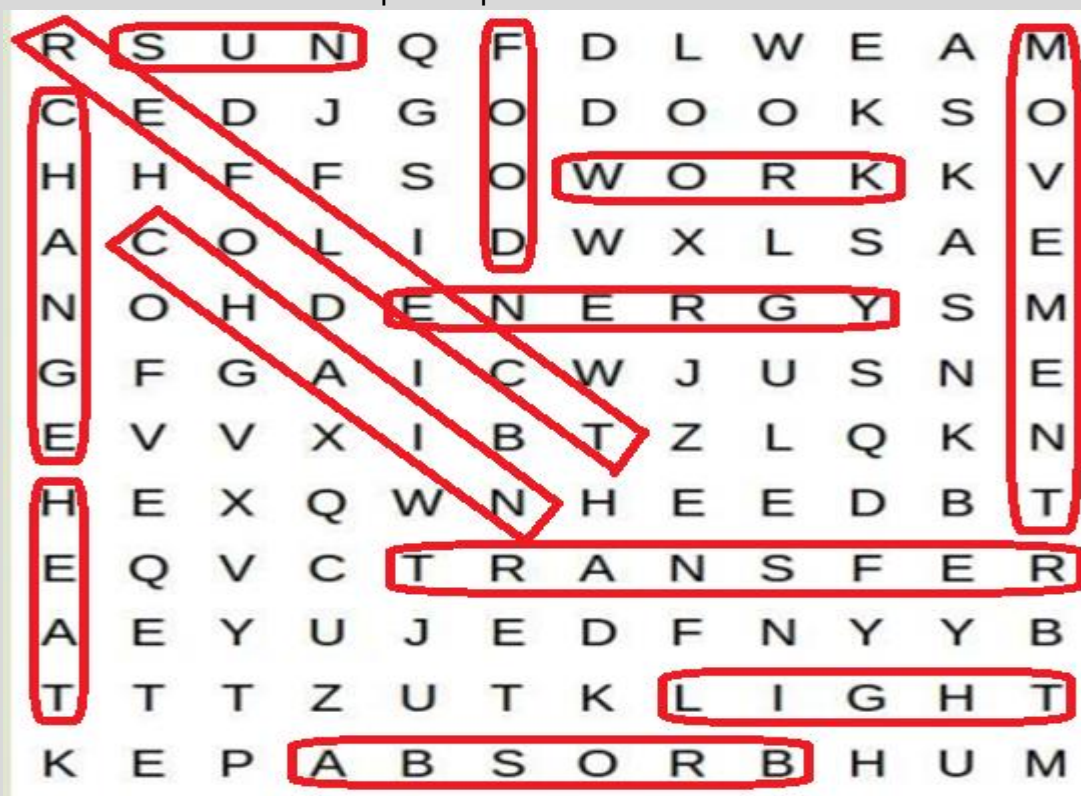
ACTIVITY: Word Search

INSTRUCTIONS:

1. Complete the word search by finding the words listed below.

Food, Energy, Work, Movement, Sun, Change, Light, Heat, Absorb, Reflect, Transfer, Chain

2. Circle them with a coloured pen or pencil.



Energy from the Sun

The main source of energy on Earth is the Sun. We call this energy **solar energy**. **Sol** means **Sun**. Without the Sun, the Earth would be a cold place with no life. Energy from the Sun has many different uses.



Rays from the Sun reach the Earth. Some are reflected and some are absorbed by the Earth

Light and warmth: We use the light from the Sun so that we can see during the day. We use the energy from the Sun to warm us.



People enjoying the Sun on the beach

Plants use sunlight to grow: Plants use the energy from the Sun to make food. As the plant grows it stores some energy in its roots, stems, and leaves.



A mealie plant

Animals eat plants for energy: The energy stored in the plants is transferred to animals for life processes.



These cows are eating grass to get their energy

Energy from wood: When we burn wood to make fire we use energy that is stored in wood. Plants get energy from the Sun to grow.



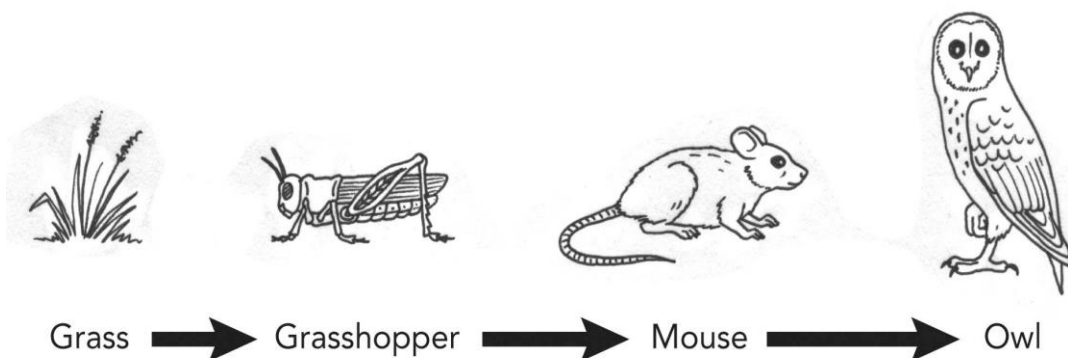
We burn wood to get warmth from the energy that is released.

Organisation of the simple food chain

The transfer of energy from plants to different kinds of animals, including people, is called a **food chain**. A food chain is a **flow of energy** in an ecosystem. Each organism forms a link in the chain as energy is passed along from one organism to the next.

The **arrows** from one organism to another show the direction of the **flow of energy**.

Look at the example of the food chain below.



Example of a food chain

ACTIVITY:

Answer the following questions:

1. What does “energy is transferred means?”
2. For each of the following appliances, decide what is the input energy and identify the output energy.

Appliance	Input energy	Output energy
Radio	Stored or electrical	Sound
hair dryer	electrical	Sound, heat energy
car moving	Stored from petrol	movement energy and heat
riding a bicycle	Stored from food	movement energy and heat
playing drums	Stored and movement	sound
lights in your home	electrical	light and heat
plants growing	light	Stored

Revision

1. Which sense organ do you use to detect sound?
 - You use your ears.
2. A hearing-impaired person cannot hear music. How do you think someone who is hearing impaired could still dance to the music?
 - They can feel the vibrations through their feet or hands. They can feel the music.
3. Which sense organ do you use to detect light?
 - You use your eyes.
4. What does "energy is transferred" mean?
 - It means energy from one part of the system to the other part of the system.
5. List three things that contain stored energy that we can use.
 - Things could be: food, wood, oil, petrol, coal, etc

2.1. Energy around us

We find energy all around us, and we use it for different purposes.

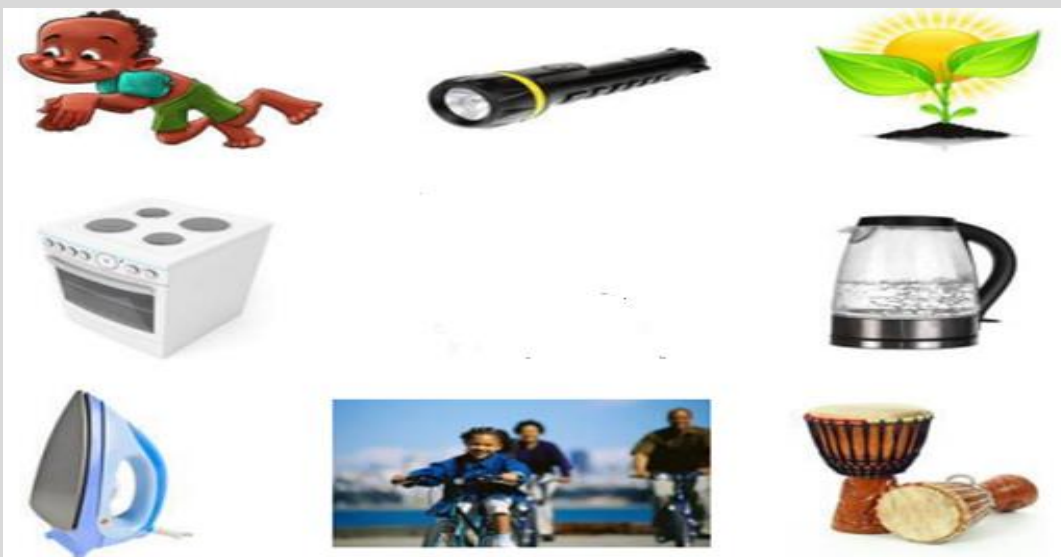
Note to the teacher: The following two activities are meant for learners to recognise how we use energy in our everyday activities. The first activity can be done with the teacher in preparation for the second one, which can be done by learners in groups. Learners might give different answers for each of the instances; the answers they give could raise interesting discussions in class.

The circled examples are only a guide, they do not reflect all the instances of energy usage.

ACTIVITY: Worksheet

For each of the pictures of objects below:

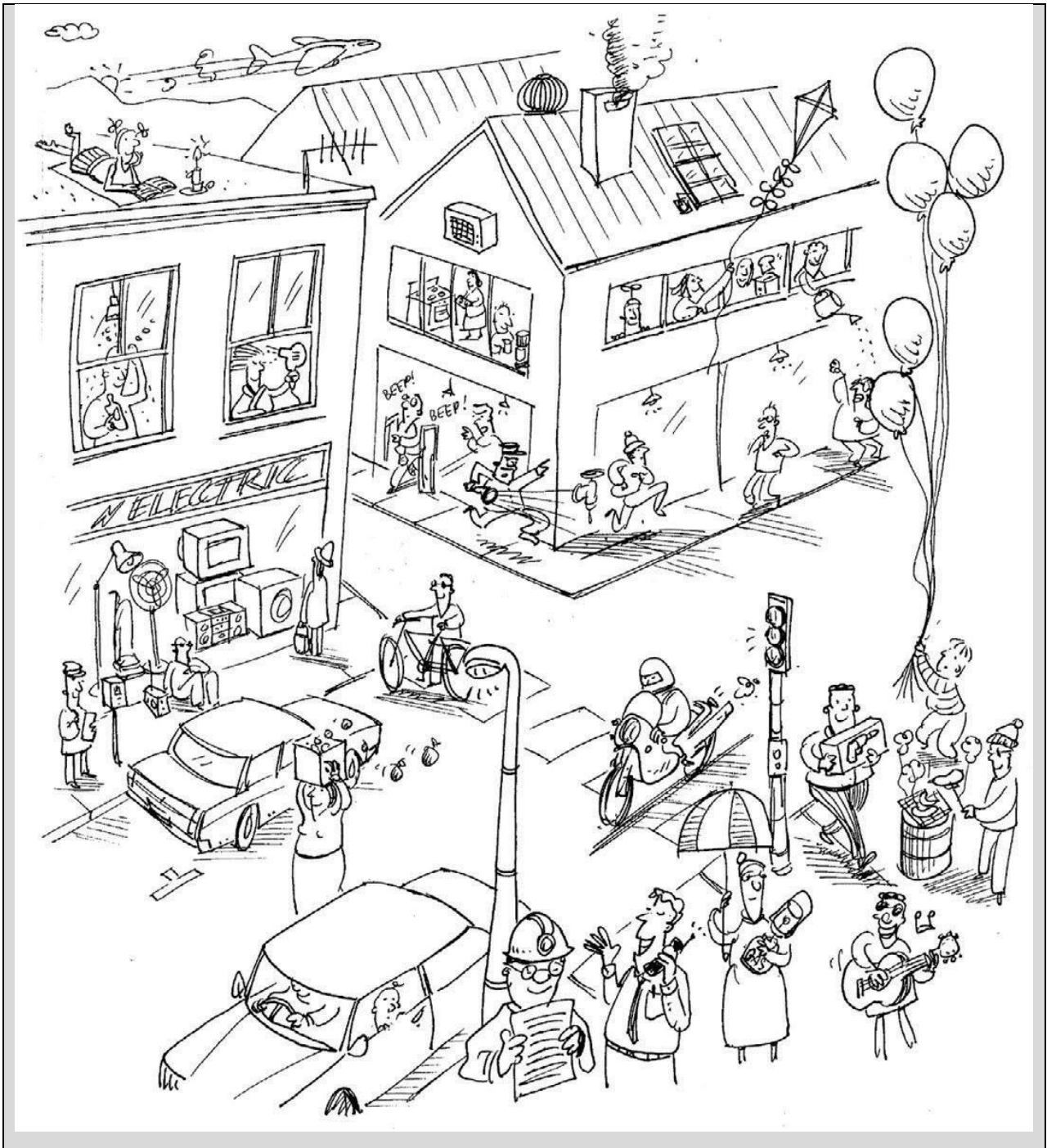
1. Indicate the name of the object, and
2. How we use energy in that object.

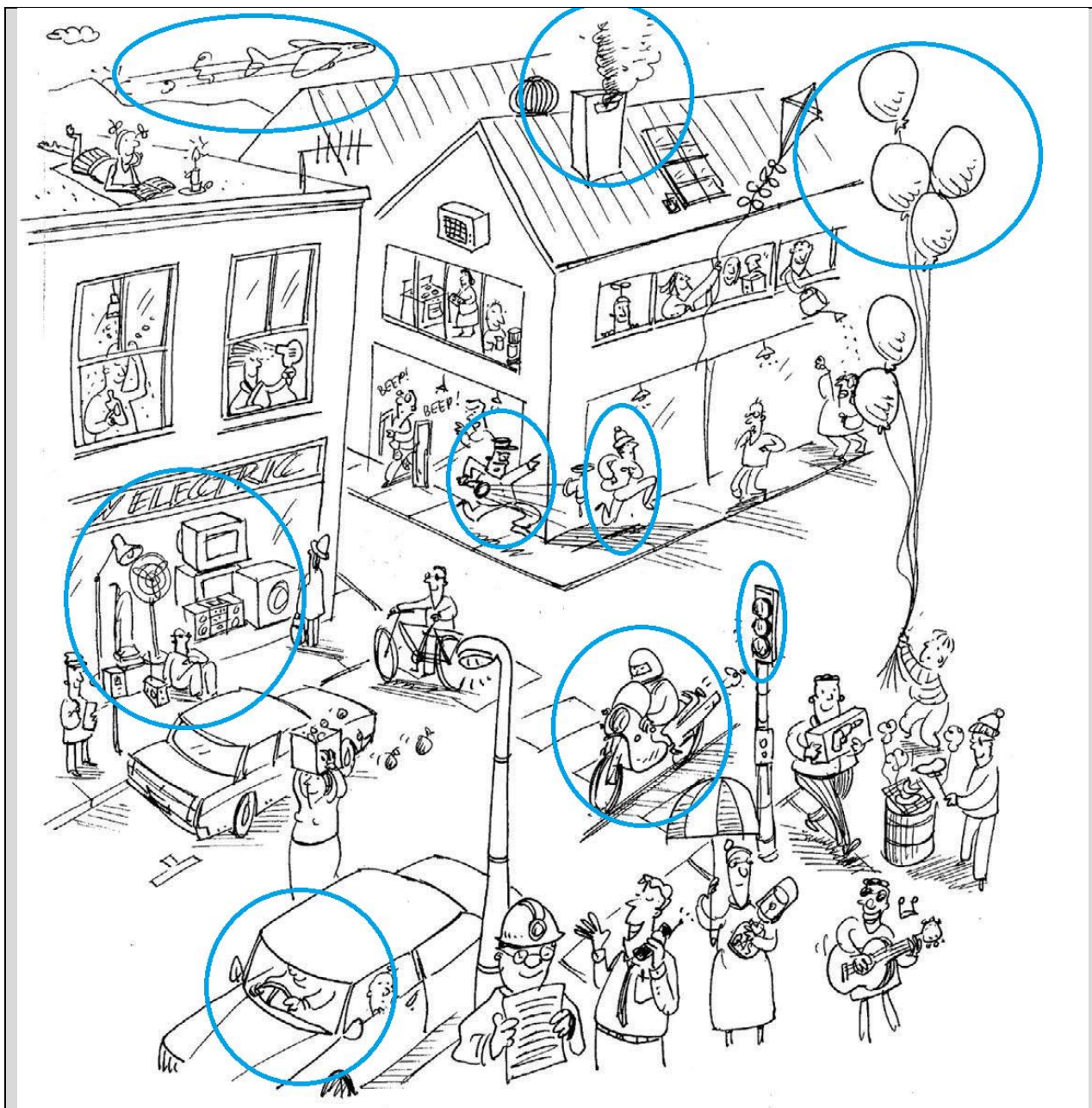


ACTIVITY: Identifying energy around us

In the diagram below:

1. Draw a circle around, or colour in all the pictures where you think energy is used.
2. Name the picture that uses energy, and
3. Indicate how people use energy in that picture.





Possible responses:

- a. Man with a Guitar – making music
- b. Traffic light – control traffic
- c. A car moving – riding/transport/moving
- d. Chimney – making fire/smoke escaping
- e. Scooter – riding/transport/delivery
- f. Man walking/running – going to work/shops/play
- g. Balloons – flying/going up

Light

Light comes from a source of light. Anything that produces light is called a source of light.

- The Sun is a source of light.
- Stars are sources of light.
- A fire is a source of light.
- A candle is a source of light.
- An electric bulb is a source of light.

Heat

Do you enjoy standing outside on a warm summer day and feeling the warmth from the Sun on your skin? What about warming your hands on a frosty cold morning in front of a fire? You are feeling heat! We discussed that the Sun provides us with light, but it also provides us with heat.



Lions lying in the Sun!

Heat can be found in many different places. Anything that provides us with heat is a source of heat.

Movement

When objects are moving, they have movement. The faster the object is moving; the more movement it has. Look at the examples of movement below.



While you are riding your bicycle, you have movement.



A race car that is travelling has lots of movement.



A rocket that is taking off has a huge amount of movement.



When I am dancing I have movement!

Stored energy

Stored energy is the energy that is stored in our food, in petrol, in wood, and in other forms. Energy is also stored in some sources. Batteries also contain chemicals which are used to store energy. This stored energy can be used for different purposes.

ACTIVITY: Stored energy

Indicate how stored energy is used in each of the pictures below:

Pictures	How stored energy is used
 <i>Energy is stored in food, such as eggs, which we eat for breakfast.</i>	When we eat food, the stored energy in the food can be used by our bodies.
 <i>Natural gas stores energy which is released as heat when it burns to cook our food.</i>	When we burn gas to cook food.
 <i>Burning coal releases the stored energy as heat and light. ¹</i>	When we burn wood or coal, the stored energy is released as heat energy that we can use to cook and to keep warm.
 <i>When your parents or taxi driver turn on the ignition in their car, the energy stored in the petrol or diesel is released.</i>	When we turn ON a car, we use the energy in the fuel to give movement energy to the car.

Transfer of energy

Energy can also be transferred (moved) from one part of a system to another part. We use electricity every day in our modern lives. Electrical energy can be transferred from a source to an appliance or light bulb.

Look at this picture of the light bulb below. Electrical energy is transferred from a source to the light bulb to make it glow.



A light bulb gives off light energy and heat energy.²



In a torch, energy is transferred from the batteries to the bulb.

Have you ever used a torch before? How do you think the bulb lights up in the torch? The batteries are the source of energy in the torch. When the torch is turned on, the energy is transferred from the batteries to the bulb to make it light up so you can see in the dark.

Sound

Sound is also a type of energy.

ACTIVITY: Observing sound energy around us

INSTRUCTIONS:

1. Close your eyes and be very quiet for two minutes.
2. Be very still and listen to your surroundings.
3. Write down five different sounds that you heard.

Sound is a special type of movement. Sound travels as vibrations that we can hear and sometimes even feel. All sounds are caused by vibration of substances.

Sound can come from different sources. You can clap your hands, stamp your feet, talk, sing or play a musical instrument. Different vibrations will make different sounds.

Note to the teacher: Discuss these conclusions from this activity with your learners.

- You cause vibrations by giving energy to the ruler.
- The ruler gets vibration energy from you.

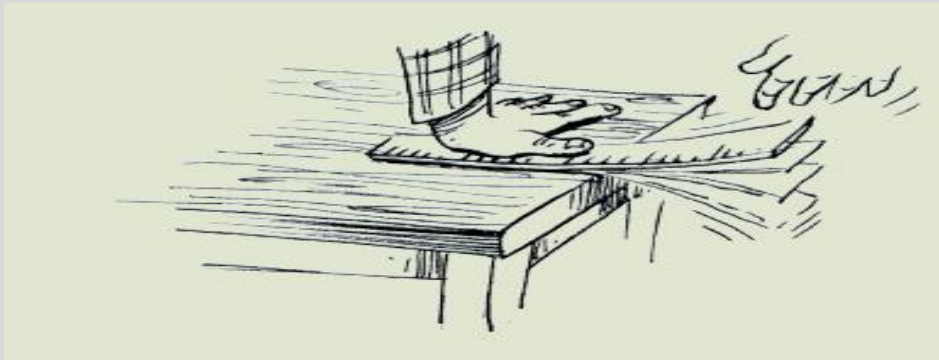
- Learners should explore sounds made from different lengths of the ruler

ACTIVITY: Use a ruler to make sound**MATERIALS**

- a ruler
- the edge of a desk

INSTRUCTIONS

1. Put the ruler on the table so that it sticks out over the edge.
2. Push the ruler down.
3. Pluck the ruler and hear the sound.

**Questions:**

1. You can see the vibrations of the ruler as it moves. Can you hear the vibrations?

Input and output energy

Note to the teacher: This activity is about “useful” and “wasted” energy in a system. A lamp, for example, is intended to provide us with light, but it can also provide us with heat that we do not need.

Learners should be able to identify “useful” and “wasted” energy in an energy system. Some learners might be interested in investigating various ways of solving the problems related to energy, such as reducing wasted energy in mechanical and electrical systems.

ACTIVITY: Investigating the input and output energy of appliances.**INSTRUCTIONS:**

Below are pictures of different appliances. Each one has input energy (electricity) and output energy which is transferred to the surroundings, such as heat, sound, light or movement.

1. Look at each image and write down the type of output energy that it transfers to the environment.
2. Some of the appliances may transfer more than one type of energy to the surroundings.

Appliances	Output energy transferred to surroundings	
	Useful energy	Wasted energy
Stove 	Heat energy	• Heat transferred to other parts of the stove • Heat warming up the room
Kettle 	Heat energy	• Sound energy
Lamp 	Light energy	• Heat energy
Vacuum cleaner 	Movement (sucking dirt)	• Sound energy • Heat energy

Electric fan 	Movement energy	• Sound energy • Heat energy
Drill 	Movement energy	• Sound energy • Heat energy

In summary, we can say that in appliances and machines (energy systems), the input is normally electrical energy and the output depends on the appliance.

Let's look at some examples

Example 1: A girl is running a race. In order for the girl to have energy, she needs to get it from somewhere. Her input energy is the **chemical energy** from the food that she ate. By running the race, she is giving out energy in the form of **movement energy** and **heat**.



Example 2: A torch will not work when you turn it on unless it has batteries. The input energy for the torch to work comes from the **chemical energy** in the batteries which is changed to **electrical energy**. The output energy from the torch is **light** and **heat energy**.



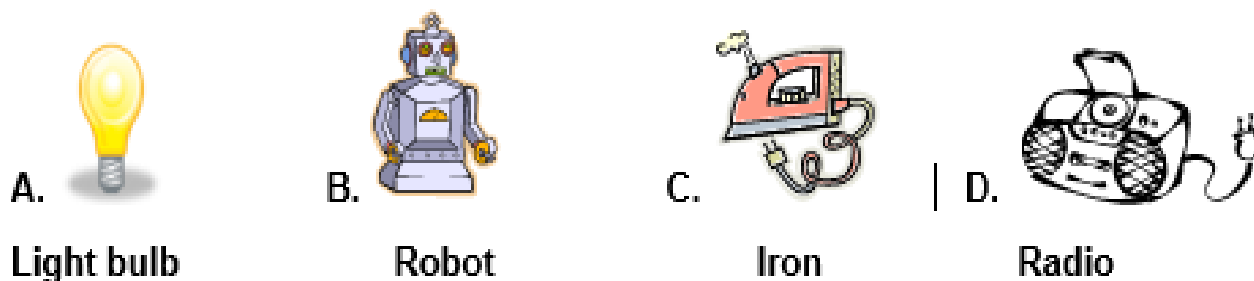
Machines and appliances

We use lots of appliances in our lives. These machines and appliances need input energy to make them work. This is usually electrical energy. The output energy (the work the appliance or machine does) is something that is useful to us.

ACTIVITY: Energy transfer

- Look at the following energy systems.
- Describe the path of energy in each system in the following way:
 - what the source of energy is (input energy);
 - the parts of the system through which the energy is transferred;
 - what do we notice as output energy?
 - what do we do with the output energy;

Describe the path of the energy transfer letter representing the picture.



Energy can be transferred to

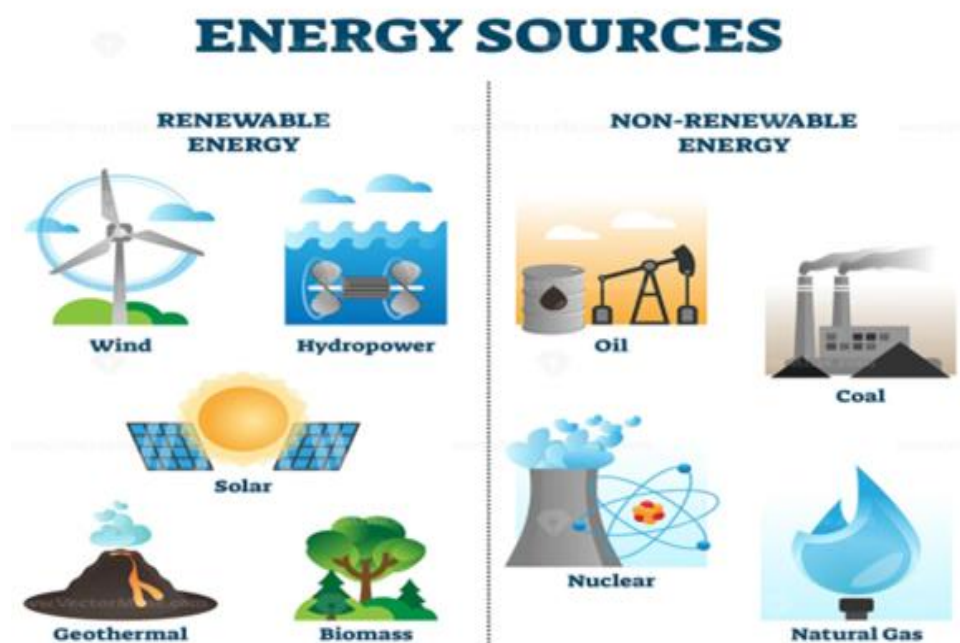
- | |
|---|
| <p>A. Light bulb: (a) electricity; (b) wires; parts of the bulb (metal parts); (c) light (d); we light up our homes;</p> |
| <p>B. Robot: (a) electricity; (b) wires; (circuit board not expected); different parts of the robot; (c) movement; (d) we get the robot to do various things;</p> |
| <p>C. Iron: (a) electricity; (b) wires; element of the iron; (c) heat/hot; (d) we iron our clothes;</p> |
| <p>D. Radio: (a) electricity; (b) wires, any part of the radio; (circuit board not expected) but speaker of the radio should be mentioned; (c) sound (music/talking/voices) (d); we dance to music; listen to the news/drama, etc.</p> |

Topic 2: Energy sources

Sources of energy

Sources of energy are things from which we produce energy. Some of these sources (things) do not get finished when used, for example, wind, water and sunlight. We therefore call wind, water and sunlight **renewable sources of energy**. Renewable sources of energy are those which can be recycled or reused.

We also produce energy from other sources that get finished when used, for example, coal, natural gas, petroleum and oil. We call these **non-renewable sources of energy**. The non-renewable sources of energy cannot be reused and so there is a limited amount available and when that runs out there will be no energy source left.

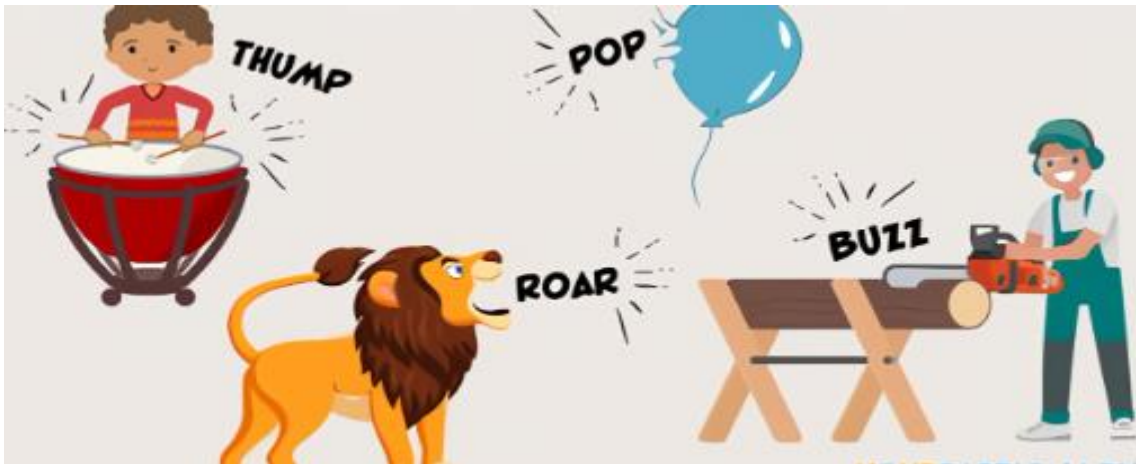


Renewable and non-renewable sources of energy

TOPIC 3: SOUND AND NOISE POLLUTION

Sound

Sound is vibrations that travel through the air or any other medium. A human ear can hear some of these vibrations but cannot hear some. For example, there are sounds that dogs can hear but humans cannot.



Noise Pollution / Sound Pollution

Noise pollution is the disturbance of the environment with very loud uncoordinated sounds that are in most cases so irritating. **Noise Pollution** disrupt regular activities and reduce quality of life.



ACTIVITY: How do we make sound when we talk or sing?

INSTRUCTIONS:

1. Rest your hand gently on your neck and hum.
2. Make low humming sounds and higher humming sounds.
3. What do you hear?
4. What do you feel?

There is a vibration in your throat when you hum which you could feel with your hand. It causes the sound you make when you speak and when you shout and when you sing.

What is vibrating in your throat? You have vocal cords in your throat. As air moves over them they vibrate. As they vibrate they cause sound.

So we make sounds when our vocal cords vibrate, but how do we hear sounds? In your ear you have eardrums. Eardrums helps us to hear the sound. Your eardrum acts

like the cling film. Your eardrums vibrate when a sound goes into your ear. This is how you hear sounds!

Examples of Noise /Sound Pollution

- Cars, buses, pedestrians, ambulances, and other street traffic noises
- Drilling or other heavy machinery in operation sounds like construction.
- Air traffic noise, such as planes taking off or landing, is always present at airports. Sounds from the workplace, which are frequently heard in open-plan offices.

ACTIVITY 11: NOISE POLLUTION



<https://www.bing.com/images>

1. List the examples of noise pollution in the picture.
 - Radio, Car, guitar, aeroplane, ambulance, Lawnmower, chain saw, speaker,
2. Name any 4 noise pollution in your area or environment
 - There may be more different examples depending on the environment.
 - Fans, generator, compressor, mills, train station, television, vacuum cleaners, hair drier, washing machine, dishwasher, etc.

Learner dependent answer

DANGER OF LOUD NOISE OR SOUND POLLUTION

Problems related to noise include **stress related illnesses, high blood pressure, speech interference, hearing loss, sleep disruption, and lost productivity.**

ACTIVITY: Making sound louder with a box

This activity can be combined with the elastic band activity earlier on.

MATERIALS:

- a cardboard shoe box or plastic box (a margarine tub or lunch box will work well)
- an elastic (rubber) band

INSTRUCTIONS:

1. Stretch the elastic between your hands and ask a friend to pluck it.
2. Now pull the elastic band over the box. The elastic must be tight. If it is not tight use a smaller elastic or a bigger box.
3. Pluck the elastic.
4. Feel the sides of the box as the elastic vibrates. Can you feel that the box is also vibrating? You may put a few rice grains inside to show the vibration of the box.



Let's summarise what we learnt from this activity:

- The box makes the sound seem louder.
- The box vibrates with the elastic.
- The vibrations of the box make sound inside the box.
- This makes the sound seem louder.
- Some musical instruments have a hollow shape.
- The sound is reflected inside the hollow.
- The walls of the hollow shape also vibrate as the sound
- echoes inside.
- This makes the sound seem louder.

ACTIVITY: Identifying noise pollution

QUESTIONS:

1. List five sounds that are very noisy, sounds that can hurt your ears, or just sounds that are difficult to live with.

- Teacher to bring 'noise making instruments' to school – pots and pans to bash together, lids of the pots and pans to bash together, repeatedly running nails over a chalkboard, playing a trumpet or other similar instrument without any knowledge of how to do this - all just M1K8N6 NO8S4! motor bike revving, loud music, cheering at sports games, aeroplane taking off, formula 1 racing, drilling holes with a drill



2. Look at the picture of a learner in a very noisy environment. Identify all the sources of noise pollution.

- People talking
- Noise from the taxi
- People shouting out the taxi
- The jackhammer on the pavement
- The aeroplane in the sky
- The exhaust and hooter from the red car
- The music blasting from the stereos in the furniture shop.

TERM 4: Planet Earth and Beyond

Knowledge Strand	Year 2	Year 3	Year 4
Planet Earth and Beyond	<u>Planet Earth</u> • Explain concepts with examples • Describe the features of the Earth • Describe the Earth as a planet in space	<u>Movements of the Earth and Planets</u> • Explain concepts with examples • Day and Night rotation • The Moon	Revision of term 1 to 3
	• The Sun • Describe the unique features of the Sun • Describe and compare the size of the Sun to that of the Earth	• Systems for looking into Space • Explain how people travel into space • Describe how telescopes are used to look into space • Describe the largest telescope build and used by South Africa	
	• The Moon • Describe the unique features of the moon		

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-4	Planet Earth	Features of the Earth • Explain the following concepts with examples: Planet, Earth, , Sun, Star, Continent, Island, Habitats,, Moon • Describe the following features of the Earth: - Shape - Size - Movement

		- Ability to support life • Explain the difference between a continent and an island. • Identify the islands, oceans and continents on Earth – with their names. • Describe the different features of habitats on the Earth. Practical: Illustrate the Earth through a model or drawings • Describe the Earth as a planet in space. • Describe how the Earth, Moon and Sun moves in space. Discuss the different ways to observe space from the Earth
5-6	The Sun	• Describe and discuss the unique features of the Sun: - Size, Movement, Distance from the Earth, Temperature, Composition, How the Sun support life on Earth
7	Moon	• Describe and discuss the unique features of the moon • Basic facts of the moon (Size, Distance from Earth etc.)
8 - 10	• The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % • Test (Pen and Paper): 60 %		

TERM 4: PLANET EARTH AND BEYOND

TOPIC 1: PLANET EARTH

Key Questions

- What is a planet?
- What are the features of planet Earth?
- What is the difference between a continent and an island?

1.1. The Earth as a planet

Note to teacher:

The definition of a planet given here is a simplified form of the new definition of a planet given by the International Astronomical Union in 2006, which is that, for a celestial body to be called a planet it must do three things:

1. It must orbit a star (in our case the star is the Sun).
2. It must be big enough to have enough gravity to force it into a spherical shape.
3. It must be big enough that its gravity cleared away any other objects of a similar size near its orbit around the Sun.

It is this new definition that disqualified Pluto from being called a planet.

*Globe must be organised and used as a model for an easy understanding of the shape of the Earth.

A planet is a very big nearly round celestial (heavenly) body that goes around the Sun on its own orbit. Sun is a star that provides heat and light energy on Earth. Planets do not make their own light; they reflect light from the Sun.

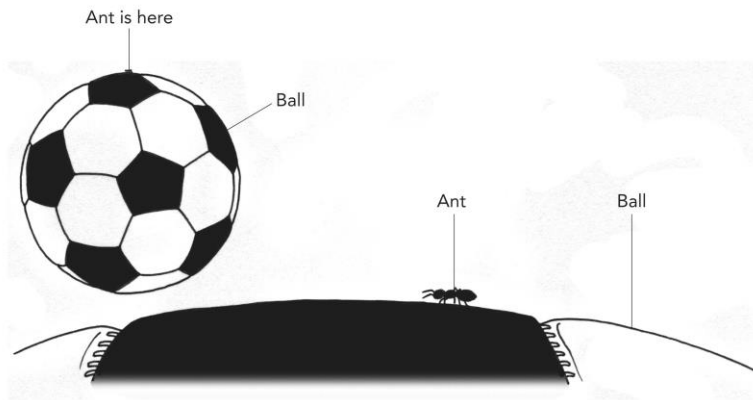
The Earth is one of the eight planets that are moving around the Sun. Earth is a ball-shaped planet travelling through space.



Planet Earth as viewed from space

The Earth looks flat, not ball-shaped. Many people long ago believed that the Earth is flat, because it looks flat. It is so big that we can't see that it is curved.

Pretend you are an ant on a soccer ball. You are so small that the ball looks flat. You can't see the other side of the ball, and you can't see that it is a ball. Look at the picture of the ant on the soccer ball. All the ant can see is a flat surface. It does not even know he is on a round ball because the ball is so much bigger than it is.



The ant on the ball sees a flat surface as it is so small compared to the ball.

This is the same as us on Earth. We are so small compared to the Earth that when we are standing on the surface, the Earth looks flat to us. We cannot see that the Earth is actually round unless we look at pictures of the Earth taken from space.

Features of the Earth's surface

Earth is a planet that supports life. The features of the Earth are the land, rocks and soil which makes up the mountains, hills and valleys, the water in dams, lakes, rivers and seas. Look at the following pictures showing the different features of the Earth's surface. The Earth is covered with a layer of air called atmosphere.



Orange river



There are lots of rocks in this landscape



*The rocky sea shore on the Garden Route A deep valley between the mountains
showing steep cliffs, rocks and the ocean*

Note to the teacher:

The activities below are meant for learners to understand how the Earth is viewed from space. Please link the activities below with view of the planet Earth from space.

Remind the learners that the Earth looks flat or level to us if we look around outside. Now ask them to imagine what the Earth would look like if they went high, very high. Many of them have seen images of the Earth as a ball, photographed from spacecraft.

They will tell you, "That is the Earth." But they may have difficulty if you ask them to show you where they are on the Earth at this moment. It is not because they do not know the continents – their problem is making the mental shift between being in Space and being on the Earth. For that reason, we spend some time in this topic developing their ability to look at things from a different point of view. This is a mental ability that normally develops when children are about 10 or 11, and as teachers we can ensure that it does develop for these learners.

The activities are arranged to provide views from lower to higher positions.

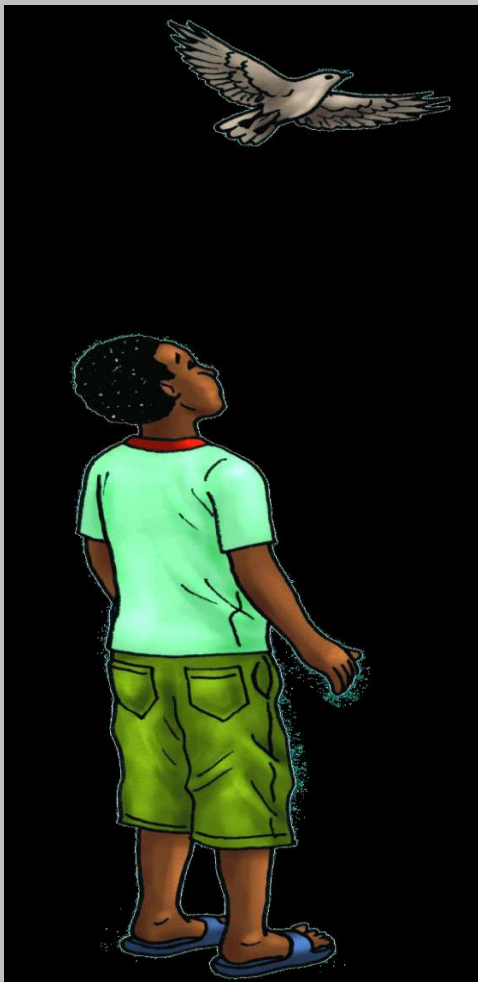
ACTIVITY: What do things look like from above?

MATERIALS:

- pencil
- coloured pencils
- rubber

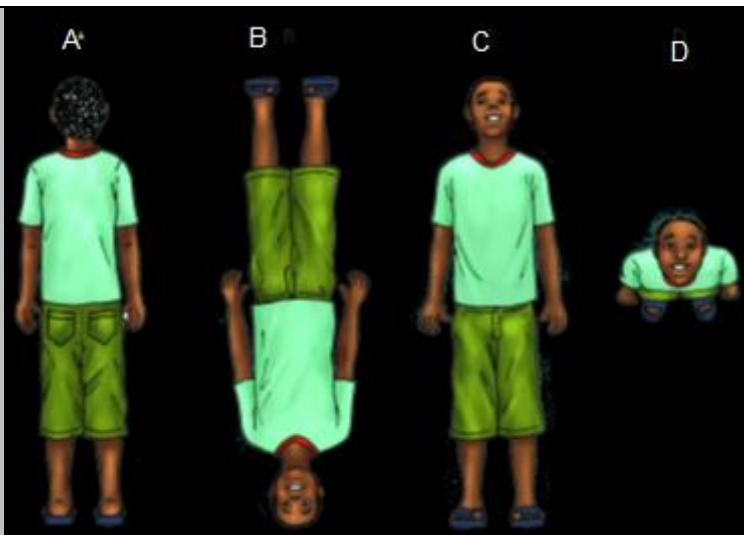
INSTRUCTIONS:

1. Let's look at what a bird sees when he flies over a boy. Can you see the bird flying over the boy in the picture below?
2. Answer the questions that follow

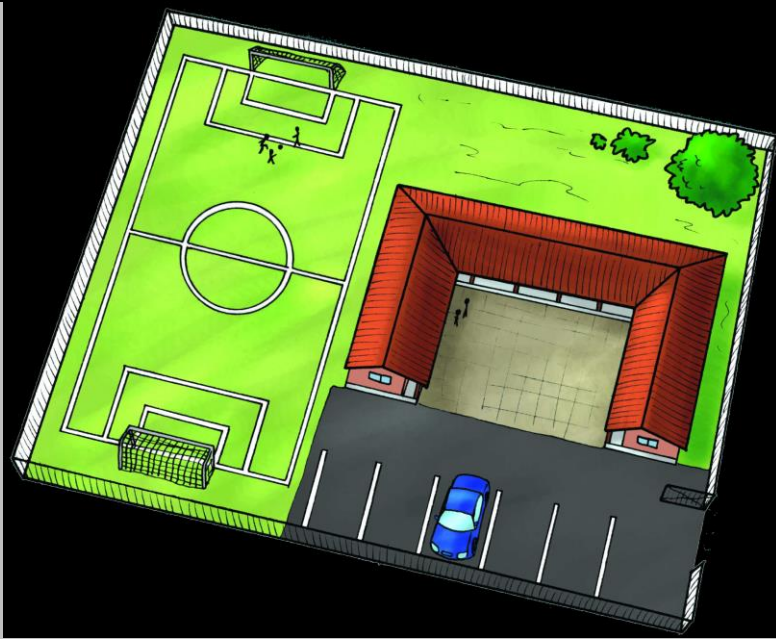


QUESTIONS:

1. When the bird looks down, what does it see? Choose the right picture from (a), (b), (c) or (d) by drawing a circle around it.



- Here you are developing the learners' ability to mentally put themselves in another place and imagine how things look from there. This is a mental ability they must have in order to understand models of the solar system.
2. Imagine you are a fly on the ceiling in the classroom. You look down and see the classroom. Now draw the classroom as the fly sees it using the space below. Draw the chalkboard, the cupboard, the door and the teacher's table. You don't have to draw the people. You can use some colour if you want to.
 - This task develops the learners' skill in visualising things as they look from a different point of view. You may find that only a few learners can imagine the features of the classroom as they look from the ceiling. For example, many learners will draw the cupboard as they can see it from the front. But they must draw the cupboard from above; they could see the doors if they were above the cupboard. Instead, they would see the things that you have put on top of the cupboard.
 3. The next picture shows you a school as it looks to a bird flying over.



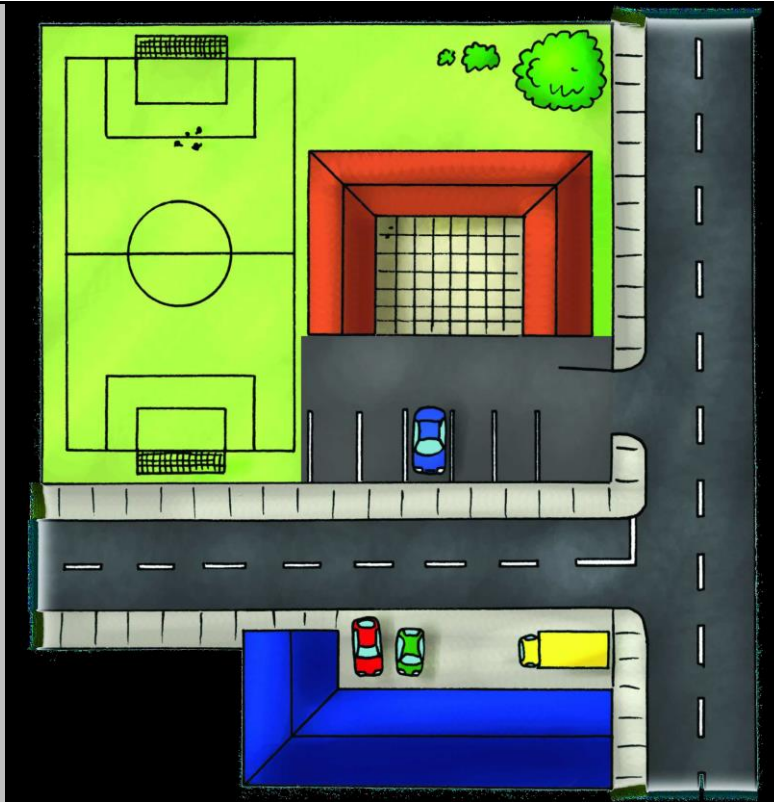
The school yard as seen by a bird

Labels have been given below for some objects in the picture.

Draw the letters A, B, C and D onto the picture to label the correct objects.

- A is the gate
- B is the roof of the school
- C is the soccer field
- D is the tree

1. An aeroplane flies over the same school. This picture shows you how the school looks from an aeroplane; the aeroplane is flying higher than the bird.



The school yard as seen by people in a low flying aeroplane

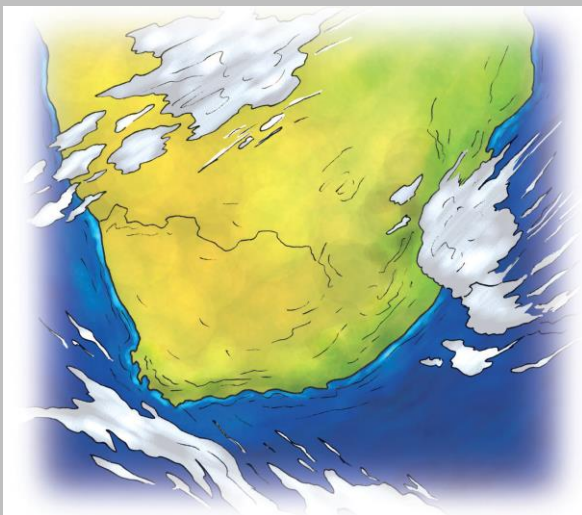
2. Find the soccer field now. Label the soccer field on the picture.
3. Why is the soccer field much smaller than in the picture where the bird is flying over the school?
 - We are much higher now, so things look smaller.
4. Find the shop. It is across the road from the school. What do you see outside the shop?
 - Answer is cars. The learners have to **make an inference** that the other building is a shop.
5. The aeroplane now flies higher up into the sky. You can now see what the town looks like to people in the aeroplane flying very high. In the picture, label the river, a road and a cloud.



The town as seen by people in an aeroplane flying high overhead

NOTE: The learners must make an inference that it is a river, road, cloud. It is not labelled. Making inferences from pictures and text is a process skill.

What we have seen in this activity is that as you go higher and higher up, the objects appear smaller and smaller. So, when photographs are taken from an aeroplane or helicopter, we get a better idea of the features of the Earth's surface and we can see more. If we go even higher up and up and up into space, then we can't go in an aeroplane anymore. Then astronauts fly up in a spacecraft. In the picture below you see what South Africa looks like to people in a spacecraft high above South Africa.



This is what people in a spacecraft see when looking at South Africa from space.



This is the Earth seen from the surface of the Moon.

1.1. The movement of the Earth

Earth moves in two different ways. It spins on its own axis; and it orbits the Sun.

Spinning of the Earth

A complete spin on the Earth's axis makes a **day** (24 hours). The axis is tilted. A day has **daytime** (when there is sunlight) and **night-time** (when there is no visible sunlight).

ACTIVITY: Who is having daytime?

What you need:

- A globe of the Earth
- The photographs of the Earth labelled Picture A and Picture B.



The Earth has an axis from the North to the South pole

INSTRUCTIONS:

1. There are two images of the Earth.
2. Look carefully at these pictures and use them and the globe to answer the questions.



Picture A



Picture B

QUESTIONS

Picture A

1. You are in South Africa. Find South Africa on the globe.
 - Learners will look at the globe and find South Africa is a country under continent Africa.
 - They must know that South Africa is in the South of Africa.
2. Find South Africa in **Picture A**.
 - They must know that South Africa is in the South of Africa.
3. Was it daytime in South Africa, when the spacecraft took the photo?

- Yes

4. Was it daytime in Saudi Arabia? Hint: Use your globe to find Saudi Arabia and then find it in **Picture A** to see if it is day or night.

- Yes, it is daytime in Saudi Arabia

5. Was it day time in Argentina when this photograph was taken? Use the globe to help you locate Argentina. Explain your answer.

- No because it was night time in South America.

6. Now use the globe to locate Brazil. In Picture A, the tip of Brazil is in the sunlight. Is it morning or afternoon in Brazil? Why?

- It is morning in Brazil because the tip of Brazil that has sunlight is in the East.

Picture B

1. Look at Picture B. What part of the Earth is this picture showing?

- West Africa and Western Europe

2. Can you see the light on in Italy? When it is dark in Italy, is it still light in Spain?

- No light in Italy and there is light in Spain

3. Is it late afternoon or morning in Spain in Picture B?

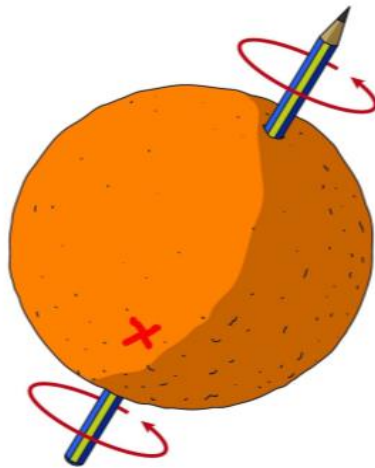
- In Spain, it is late afternoon

Imagine an orange with a pencil stuck through it as in the picture below. The pencil is the axis of the orange. The Earth does not really have a pencil through it, but it does spin around like this orange on the pencil.

Note to the teacher:

The following activity is intended for learners to do it practically.

- They will use various materials to create a model of the Earth as below.
- Learners should also model the movement of the Earth around the Sun. The Earth spins as it moves around the Sun.



*The Earth is like the orange and the pencil is like the axis.
The curved arrows show which way the Earth spins.*

We are on the Earth. Let us imagine we are at the point where you see the red "X" on the orange:

- The Sun shines on the Earth and so we, at **X**, see the Sun. We call that daytime.
- But the Earth never stops spinning. So we, at **X**, move around with the Earth into the shaded part of the Earth. Then we cannot see the Sun any longer then it is night-time for us at **X**.

Orbiting of the Earth

As the Earth spins, it also orbits (moves around) the Sun. One complete orbit of the Earth is called a **year**. By the time the Earth completes its orbit around the Sun, it has completed $365\frac{1}{4}$ spins (days). We say a year is $365\frac{1}{4}$ days. The quarter day occurs because the Earth completes the orbit before completing a day.

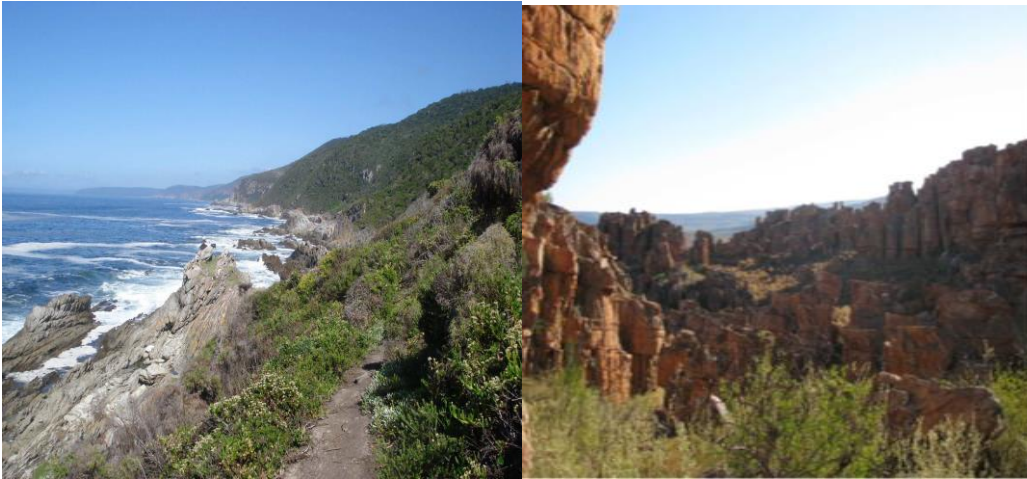
In Space, the Earth keeps on going around the Sun as it spins.

1.2. How Earth supports life

Planet Earth supports life because of favourable temperatures, availability of water, soil and air.

1.3. Habitats

A habitat is a place where living organisms (plants and animals) live so they can find food, water, shelter and have their babies. The Earth has many habitats for many different living things. The different features of the Earth above are a few examples that provides habitats for various plants and animals.



The rocky sea shore on the Garden Route There are lots of rocks in this landscape showing steep cliffs, rocks and the ocean

ACTIVITY: Habitats on Earth

INSTRUCTIONS:

- Look at the pictures again on the pages before this activity showing different features of the Earth's surface.
- Answer the questions below.

QUESTIONS:

1. Name some of the living things you can find on Earth.
 - *Plants and animals is the short answer. But make sure the learners give you plenty of examples. Examples are trees, bushes, grasses, birds in the trees, insects that the birds eat, goats, etc. Build up the idea of living things here, because you will have to teach how the Sun provides light and warmth for them.*
2. In the pictures, where do the birds live?
 - *In the trees, on the beach and river-banks, some float on the water and dive for fish.*
3. In the pictures, where do fish live?
 - *In the river and in the sea*
 - Where can cattle live?
 - *Along the banks of the river, on the grasslands*
4. What kind of animals can live in deserts? A desert is a place with almost no water.
 - *Snakes, bats, meerkats, and jackals are examples.*
5. What kinds of animals live in forests?
 - *buffaloes, elephants, bushpigs, and monkeys are examples.*
6. Habitats have unique features, such as the rocky shore habitat which has crashing waves and big rocks. Name four habitats that you have seen in the pictures.

- *rivers, seas, grasslands, forests, mountains*

7. Name four non-living things you can see in the picture.

- *The features like air, clouds and rivers are all non-living things. Many learners won't believe this; for example, they consider that a river and a cloud are living. Remind them of what you did in Term 1 for Life and Living.*

1.4. Continents

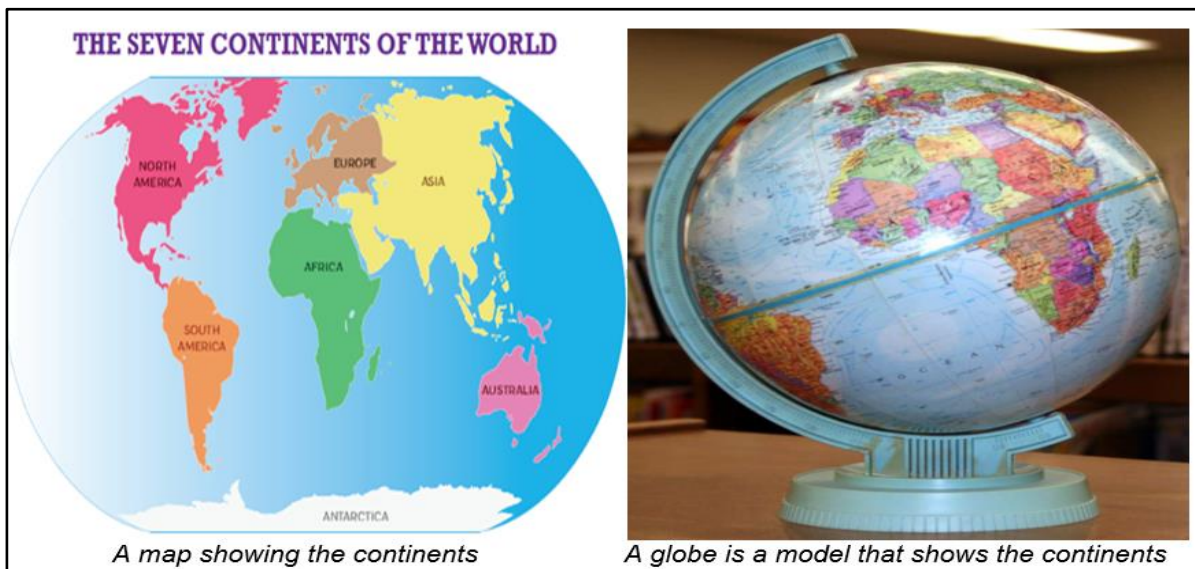
Note to the teacher:

Learners should be shown all the continents on a map or globe

A continent is one big piece of land on the Earth. A continent has many countries. Africa is a continent with more than 50 countries, and South Africa is one of them.

The Earth has seven continents; they are:

1. Africa
2. North America
3. South America
4. Asia
5. Europe
6. Australia
7. Antarctica



1.5. Island

Note to teacher

Learners should be shown some islands on a map or globe, especially those mentioned here, Madagascar, Zanzibar, Cape Verde and Mauritius, as well as

others so they do not think those are the only islands. They can also be asked to identify some islands themselves.

An island is a piece of land that is completely surrounded by water. Robben Island is a small island in South Africa. Madagascar, Zanzibar, Cape Verde and Mauritius are some of the African countries that are islands.

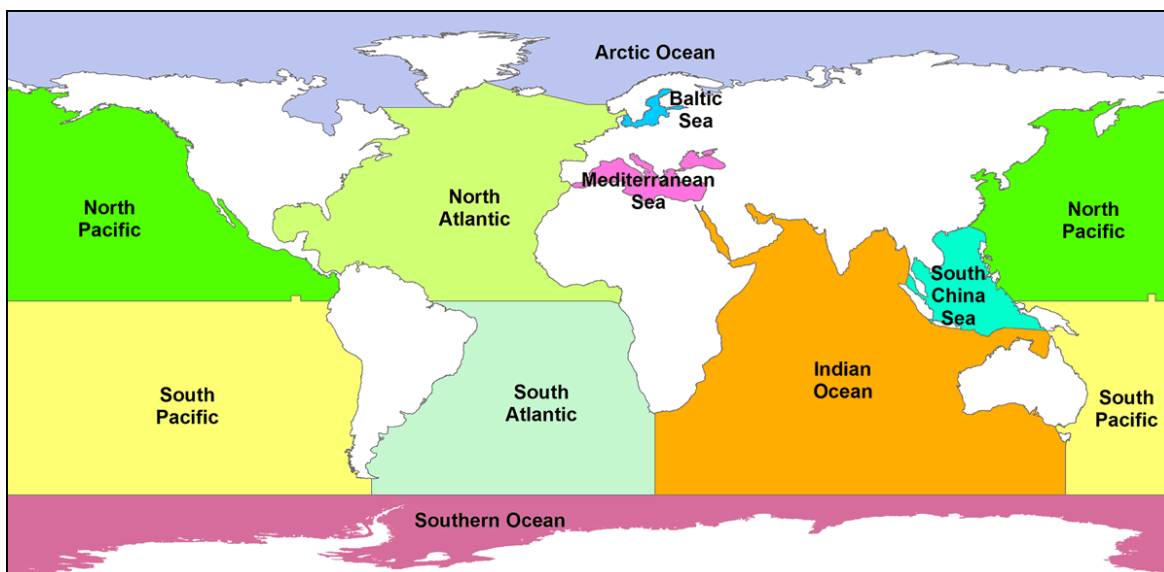
1.5.1. Difference between the Continent and Island

A continent is a big piece of land made up of many countries. An island is a small piece of land that is surrounded by water. It is either part of a country or only one country.

1.6. Oceans and Seas

Most of the surface of the Earth is covered with water (oceans and seas). This is why it is called the Blue Planet. Many people think that an 'ocean' and 'sea' is the same thing.

An ocean is a very large mass of water which covers a huge part of the Earth's surface. A sea is much smaller than the ocean and a sea is normally surrounded by land on some sides.



Oceans and seas of the world

Summary

- A planet is a very big nearly round celestial (heavenly) body that goes around the Sun.
- The Earth is one of the eight planets orbiting (moving around) the Sun.
- The features of the Earth provide different habitats.
- Big areas of land are called continents, and areas of land with water all around them are called islands.
- The Earth spins on its axis and this is the reason we have a day.

- If we go up high above the Earth, everything looks different to the way it does when we are on the ground.

REVISION: PLANET EARTH

1. A person's nose and eyes and mouth are features of his or her face. Name four features of the Earth.
 - Land (mountains, hills, valleys), water (oceans, seas, rivers, lakes), air (clouds).
2. The earth is shaped like a ball. People do not fall down off the ball. What is the reason for that?
 - The down direction is the direction that points to the centre of the Earth. Gravity pulls everyone towards the centre of the Earth.
3. List the seven continents of the Earth.
 - Africa
 - Antarctica
 - Asia
 - Europe
 - North America
 - South America
 - Australia
4. Name an island that is part of South Africa.
 - Robben Island, Dassen Island, Seal Island

NOTE: Let the learners use whatever language they are most fluent in, and then teach the English words later. Their tables should then have both the mother-tongue and the English words in them.

5. Complete these sentences. Write out the whole sentence on the open lines, and use some of the words in the word-box to complete your sentence.

Word box: • water • continent • Blue Planet • one part • clouds

- a. The planet Earth looks blue and white from space. It is called the because Blue planet it is covered with water and clouds
 - b. A continent is a large piece of land. For example, Africa is a Continent; and South Africa is just one part of Africa.
6. How many hours are there in a day?
 - 24 hours
 7. How many hours pass from sunrise until the next sunrise?
 - 24 hours

8. How many days pass between your 10th birthday and your 11th birthday?

- 365¹/₄ days

9. How many times must the Earth spin around between your birthdays?

- Earth 365¹/₄ days

10. Write out the whole paragraph and complete it using some of the words/phrases in the word box. You do not need all the words/phrases.

TOPIC 2: THE SUN

Key Questions

- What are the features of the Sun?
- How far away is the Sun from the Earth?
- The Sun looks so small in the sky, so how can it be bigger than the Earth?
- Why is the Sun so important to us and all living things on Earth?

.1. The Sun as a star

Note to the teacher

The Sun is a star, and it is the only star in the Solar System, i.e. *system of the Sun*. This makes the Sun the closest star to Earth. The other stars we see in the sky are all very far and outside of the Solar System.

We see many stars at night. Many of those stars are very much bigger than the Sun. They look small because they are so very far away. Stars are balls of very hot gas and they make their own light. We can see many more stars through a telescope



Hundreds of thousands of stars seen through the Hubble telescope

Stars that look orange-reddish are not as hot as the Sun, and stars that look blue-white are much hotter than the Sun.

The Sun is the only star in our Solar System. Solar means Sun, i.e. the system of the Sun. The Solar System consists of the Sun, at the centre, eight planets, and other objects that move around it.

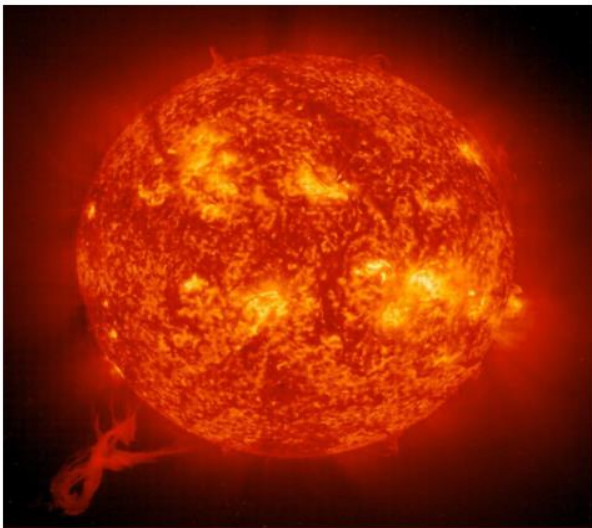
The Sun is a brightest star that is not moving while the planets move around the Sun in their orbits, as shown in the diagram below. The Sun and all the planets pull on each other as the planets move around it.

The Sun is the closest star

Why does the Sun look so much bigger than the other twinkling stars?

This is because the Sun is the closest star to us on Earth. The other stars in the sky are much, much further away.

The Sun is a very big ball of hydrogen gas that is changing into helium gas, all the time, and this change gives off energy which makes the Sun very hot. It is nearly 150 million kilometres away from the Earth. The Sun is extremely hot - nearly 15 million degrees Celsius at the centre and 5 500 degrees Celsius on the surface.



*The Sun looks like this through a special camera.
Never look at the Sun with just your eyes.*

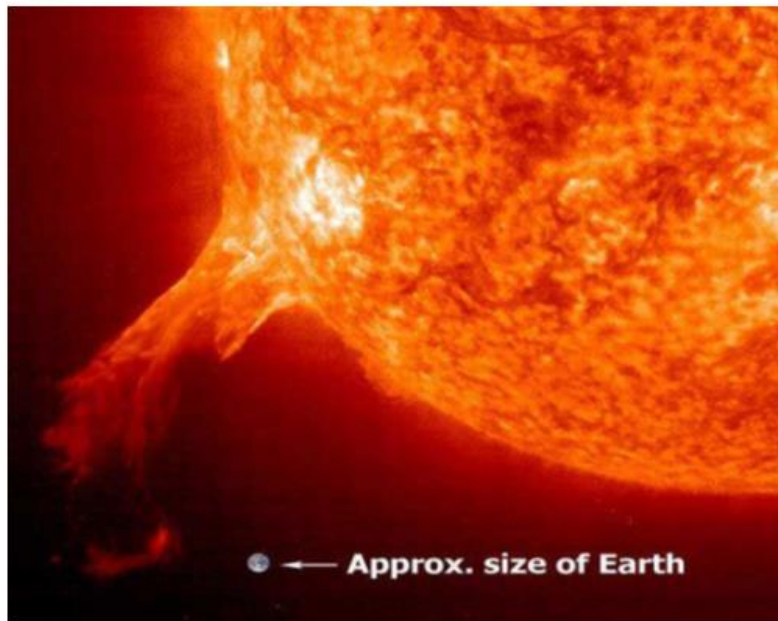
Note to the teacher:

The following activity can be done outside on a pavement where you can draw with the chalk, or else in a sandy area where you can just draw with your finger. The pavement would work better.

.1.1. The Sun is very much bigger than the Earth

The Sun may look smaller than the Earth up in the sky. But this is actually because it is very, very far away. The Sun is much bigger than the Earth, it is so big that 1 300 000

Earth can fit inside the Sun. The Sun is way bigger than the Earth. The Sun's diameter is 1 392 000 kilometres and the Earth's diameter is 12 756 kilometres.



*This is how the size of the Earth compares to the size of the Sun.
The Earth is not really this close to the Sun.*

ACTIVITY: How big is the Sun, compared to the Earth?

MATERIALS:

- a piece of pavement
- a ruler or tape measure
- a piece of string longer than 60 cm
- a piece of chalk

INSTRUCTIONS:

1. Draw a circle 1 cm in diameter on the pavement with the piece of chalk. This represents the Earth.
2. Now, move about 1,5 to 2 metres away from the little Earth and give yourself enough space to draw the Sun.
3. Draw a circle 108 cm in diameter – this represents the Sun.
4. **To draw a circle with a diameter of 108 cm**, tie a string around the chalk. Measure 54 cm of string from the chalk and tie a knot there. Hold the knot on the board and move the chalk around the knot to draw a circle.

QUESTIONS:

1. If the Sun is so much bigger than the Earth, why does it look so small to us?

The Sun looks small because it is so very far away.

The pavement drawing shows the Earth quite close to the Sun. The Earth is actually very far from the Sun. It is 150 million kilometres from the Earth. That is 150 000 000 kilometres.

That is a really long way from the Earth to the Sun. If you went in a car at freeway speed of 120 km/h, you would have to travel for 146 years to reach the Sun. So, the Sun is far away and it is very big and very hot.

If the Sun is so far away, why does it look as big as it does? The Sun is so big that it is difficult for us to understand how big it really is. A model can help us to understand.

ACTIVITY: Use a model to show how far the Earth is from the Sun.

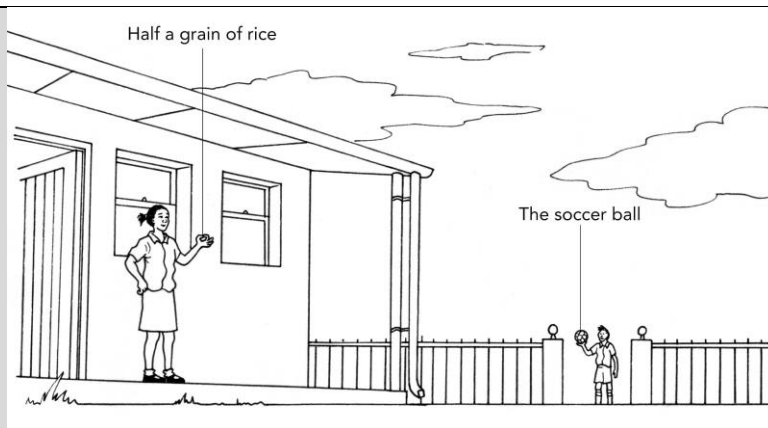
MATERIALS:

- a grain of rice broken in half
- a soccer ball
- space to move, such as the playground or soccer field

INSTRUCTIONS:

1. Look at the picture below.
2. One learner stands in a place where there is a lot of room all around them. This learner holds the soccer ball; this ball represents the Sun.
3. Another learner stands next to her and holds the half-grain of rice. It represents the Earth.
4. The learner holding the rice walks away from the learner holding the soccer ball with 24 of the biggest steps he can make. That distance is about 24 metres. The 24 metres represents the distance from the Sun to the Earth.
5. Now the learner carries the half-grain of rice, walking to walk to his right side. He must always stay 24 metres from the soccer ball. If he does this, he will walk in a circle around the soccer ball.

This model shows us that the small Earth moves in a circle around the big Sun.



The soccer ball represents the Sun, and the half-grain of rice represents the Earth.

QUESTIONS:

1. Stand 24 metres away from the soccer ball. Hold up one finger in front of you, and cover the soccer ball with your nail. Is the soccer ball really as big as your fingernail?
 - No
2. Why does the ball look as big as your nail?
 - The ball looks smaller because it is so far away

2.2.3. The Sun is important to life on Earth

Without the Sun, life on Earth would not be possible. It would be completely dark and freezing cold. The Sun provides the Earth with light and heat which are used for life on Earth.

The Earth is the only planet in our Solar System with the right temperature for us to live; it is not too hot and not too cold. It is at exactly the right distance from the Sun, and has the suitable atmosphere for the perfect temperature to support life.

REVISION: THE SUN

3. Write out these sentences and complete them. Choose some of the words from the word box to complete the sentences. Write out the whole sentence.

Word box: (You do not need to use all the words)

- hydrogen gas
- helium gas
- plants
- light
- heat
- half a grain of rice
- Sun
- a soccer ball
- the Moon

When we compare the size of the Earth to the size of the _____; the Earth is the size of _____ compared to the size of a soccer ball.

The Sun gives _____ and _____ to the Earth.

All _____ need light and heat.

The Sun is not burning like a wood fire. The Sun is hot because _____ is changing into; _____.

Answers

When we compare the size of the Earth to the size of the Sun; the Earth is the size of half a grain of rice compared to the size of a soccer ball.

The Sun gives light and heat to the Earth. All plants need light and heat.

The Sun is not burning like a wood fire. The Sun is hot because hydrogen gas is changing into; helium gas

TOPIC 3: OUR MOON

Key Questions

- Why is our Moon bright on some nights?
- What is the surface of our Moon like?

3.1. Features of our Moon

Our Moon is a rocky body orbiting planet Earth. It is smaller than the Earth, our Moon's surface is made of rock and grey sand. Its surface is also covered with holes called

craters. Craters are large bowl-shaped holes on the ground. They are made by rocks from Space that hit the Moon.

ACTIVITY: Activity: Making a crater

Materials:

- Moist soil
- Wash basin/big open container
- Different sizes of stones

INSTRUCTIONS:

1. Fill the container with moist soil.
2. Drop different sized stones onto the soil in the container, one at a time.
3. Carefully remove each stone from the soil.
4. Observe what is left on the soil.

QUESTIONS:

1. Describe what you see.

Learners will observe different sized holes

2. What do the holes represent?

Craters

3.2. Movement of our Moon

It is not a planet but a natural satellite of the Earth. A satellite (follower) is a body that orbits around another body in Space. The average distance between the Earth and our Moon is 384 400km. Like the planets, it produces no light of its own, but we can see it because it reflects the light from the Sun.

Our Moon rotates on its own axis, the same as its orbital period around the Earth and that is the reason we always see the same side of the Moon. The moon has no atmosphere, so there is no air to breathe, no wind to blow the dust, no weather and no water, so there is no rain to wash away sand. That is the reason why we see the craters on the Moon so clearly.

The Moon is the only place in our Solar System, other than the Earth, where humans have been. Moon can be viewed through binoculars. We see the moon both during the day and at night.



The Moon

QUESTIONS

2. What made the craters on the surface of the Moon?
 - Rocks that travel through Space, called meteoroids. They crash into the Moon at very high speeds.
3. The surface of the Moon has many craters but the surface of the Earth has very few craters. Explain why that is so.
 - The Moon has no atmosphere to protect it. Earth has an atmosphere so most rocks from space burn up before they reach the ground.
 - (b) Earth has been hit by rocks, as the Moon was, but erosion has worn down the craters so that we cannot see them clearly any more.
4. Neil Armstrong was the first man to put his foot on the Moon. His footprint is still there after 40 years. On Earth, a footprint does not last so long. Explain why it lasts so long on the Moon.
 - The Moon has no rain and no wind to remove the footprint

REVISION: The Moon, Earth and the Sun

1. Complete the sentences using all the words from the word box. Write the sentences out in full on the lines below and number each sentence.
 - a) The Sun stays in _____.
 - b) The Earth rotates on its own _____.
 - c) The Moon revolves _____.
 - d) The _____ revolve together in a big circle around the Sun.
 - e) We only ever see _____ of the Moon.

• Moon and the Earth • around the Earth • one side • axis • the same place

Answers

- a) The Sun stays in **the same place**.
- b) The Earth rotates on its own **axis**.
- c) The Moon revolves **around the Earth**.
- d) The **Moon and the earth move** together in a big circle around the Sun.
- e) We only ever see **one side** of the Moon.